Synthetic X-Ray Bag Scan Generation System

Page Documentation

Pages Covered:	Upload, Library, Collage, Saved, Load
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Comprehensive documentation of all five application pages including features, workflows, edge cases, backend behavior, and technical implementation details.

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1 Introduction

1.1 README Overview

The Synthetic X-Ray Bag Scan Generation System is a full-stack application for generating and reviewing synthetic X-ray bag scan data.

- **Frontend:** React + Vite + TypeScript + Tailwind CSS
- **Backend:** Express + file processing pipelines
- **Python scripts:** NumPy / SciPy / OpenCV / Pillow for raw pixel processing

1.2 Quick Start

1. Clone and enter the repo
2. Configure environment (defaults are safe)
3. Run via Docker Compose or Node.js
4. Open **http://localhost:5174**

1.3 Pages Documented

This report covers five application pages:

1. **Upload Page** — Upload and process pixel data files
2. **Library Page** — Manage library items (bags, threats, non-threats)
3. **Collage Page** — Generate synthetic X-ray combinations
4. **Saved Page** — Manage saved layout combinations
5. **Load Page** — Upload annotations, check library availability, and generate combination images

2 Upload Page

2.1 Page Layout

The Upload Page is the entry point for adding new X-ray scan data to the system. It accepts raw scanner files (`.dump.pgm`, `.pgm`, `.tip`, `.dump`) and pre-extracted pixel bundles (`.npz`). `.npz` files already contain the high- and low-energy pixel arrays, so they skip the raw-scanner decoding step and are split directly into `highenergy/<id>.npz` and `lowenergy/<id>.npz` with a preview PNG generated from the high-energy channel.

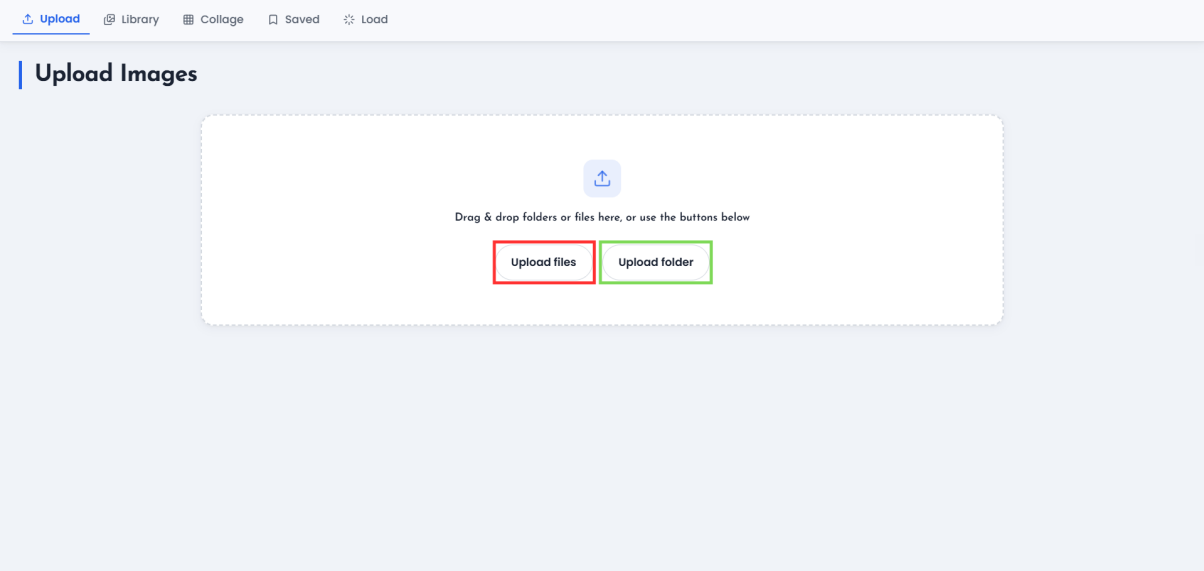


Figure 1: Starting Upload Page

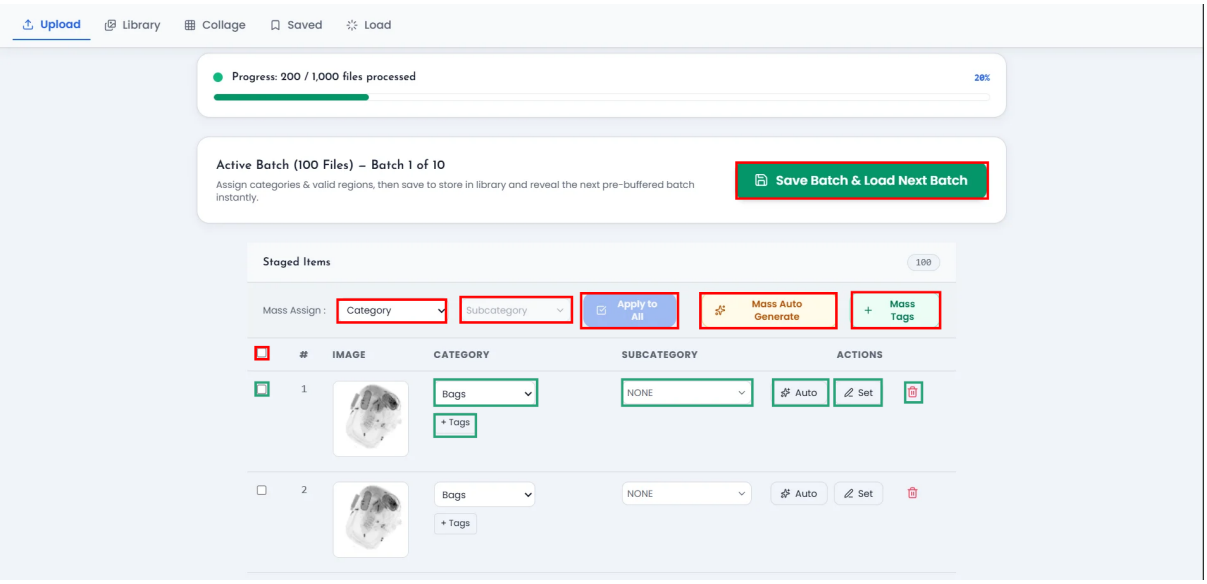


Figure 2: Upload Region with Buffer

When 200 files are loaded, only 100 are shown in the staged item table while the remaining 100 are held in an internal buffer. The buffer pre-processes the next batch so that when the user saves or navigates to the next batch, there is no delay.

Each row in the staged items table contains per-item controls, highlighted in green in the screenshot: Select, Category, Subcategory, Auto Valid Region, Manual Valid Region, Delete, and Tags.

The red-highlighted section at the top of the table is the mass action bar. It provides the same capabilities as the individual controls but applies them to all 100 staged items at once, sparing the user from repeating each action 100 times. The available mass actions are Mass Assign, Mass Auto Generate, and Mass Tags.

Clicking Save Batch persists the current 100 items to the library, after which the next 100 items are automatically loaded from the internal buffer into the table.

- **Singular category** — Per-item category dropdown
- **Singular subcategory** — Per-item subcategory dropdown
- **Singular auto** — Per-item auto-generate valid region button
- **Valid region** — Status indicator showing whether a region exists
- **Editor button** — Opens the full-screen valid region editor
- **Delete button** — Removes the item from the staging table

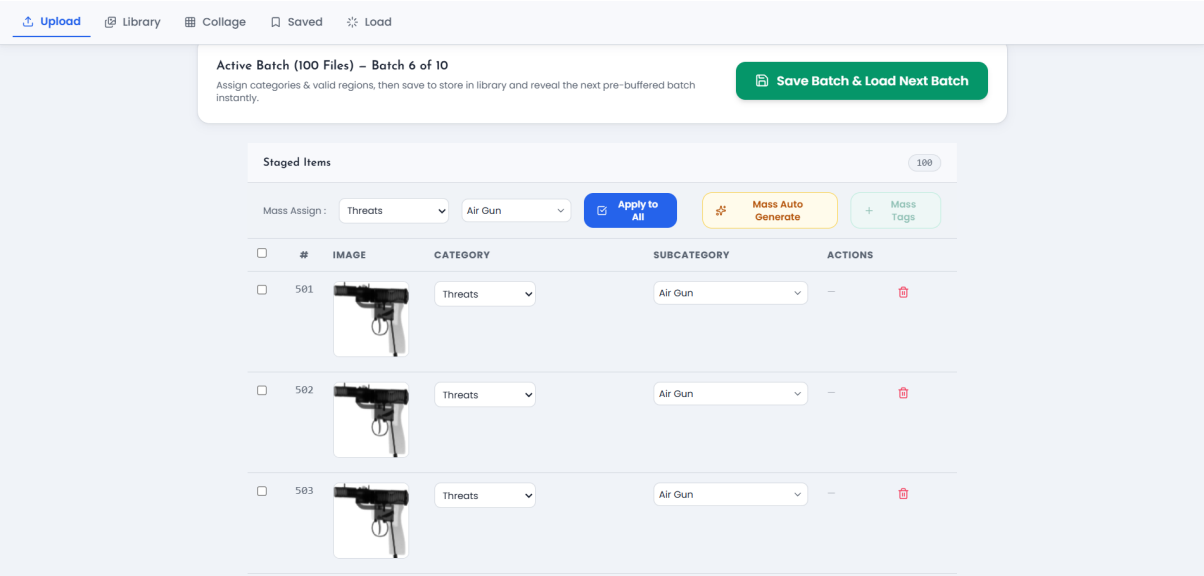


Figure 3: Tip File Upload

When uploading `.tip` files, the system processes each file into pixel data (PNG + `.npz`) and registers the item with the appropriate category (threats in this case). The exact same process applies for threats and non-threats — the only difference is the category assigned during staging.

When uploading `.npz` files, the system skips the raw-scanner decoding step because the pixel data is already present. The high- and low-energy arrays are extracted directly, separated into `highenergy/<id>.npz` and `lowenergy/<id>.npz`, and a preview PNG is generated from the high-energy channel. The item is then registered in the library with the user-assigned category, exactly like any other upload.

2.2 Category and Subcategory Selection

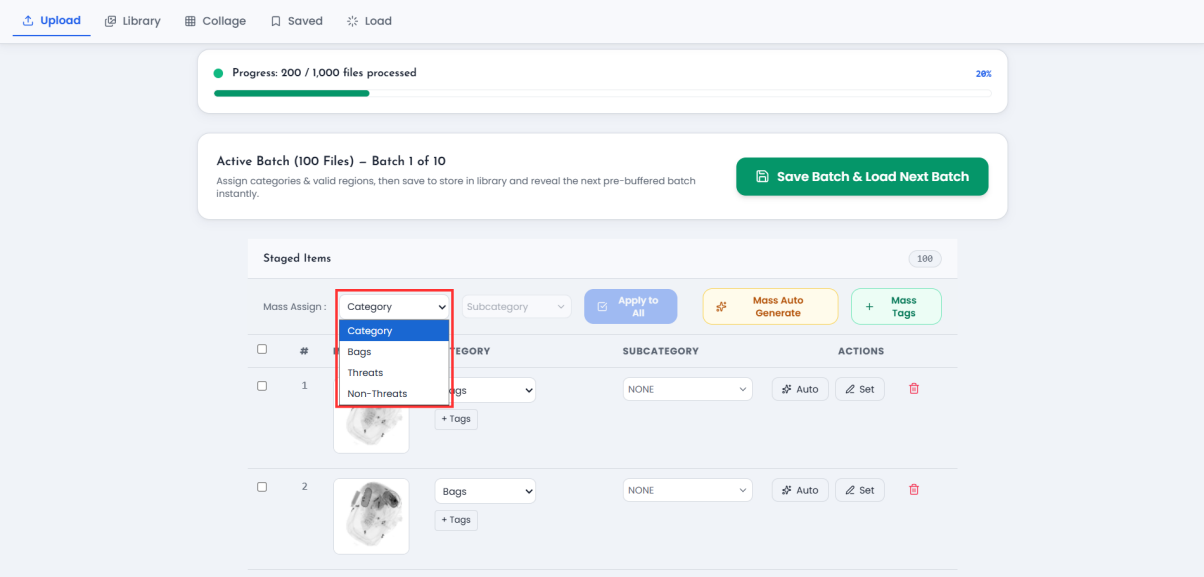


Figure 4: Category Dropdown

Each staged item has a category dropdown that shows three options: **bags**, **threats**, and **non-threats**. Selecting a category assigns it to the item.

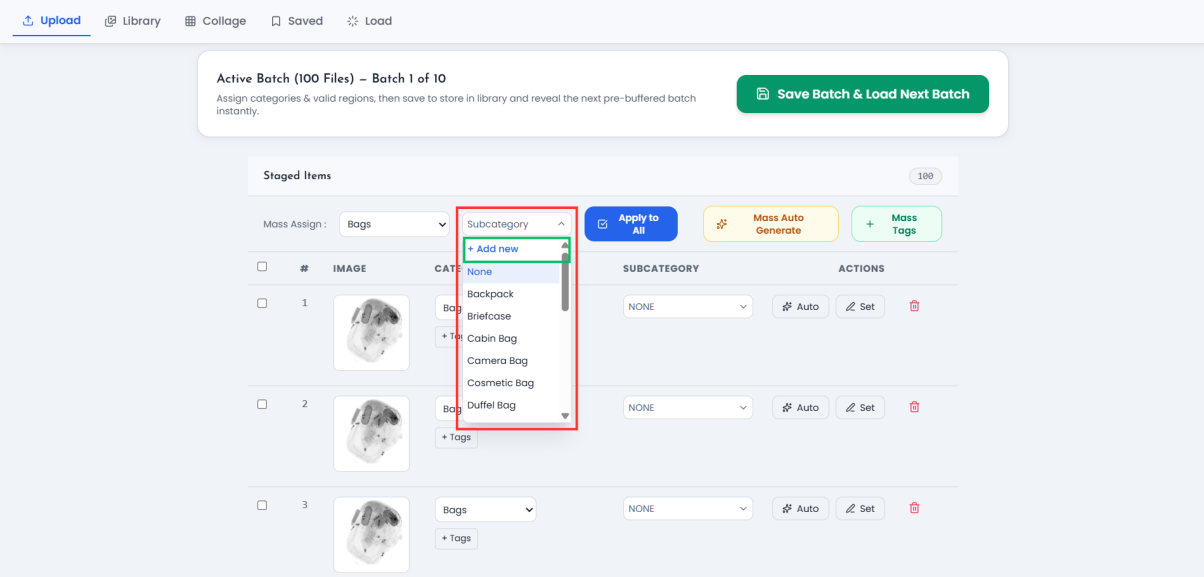


Figure 5: Subcategory Dropdown

Beside the category dropdown, the subcategory dropdown lists all existing subcategories for the selected category. An **Add New** button at the top lets the user create a new subcategory on the fly.

When the user clicks **Add New**, a text input box appears directly in the dropdown. The user types the new subcategory name and presses Enter. The new subcategory is immediately saved to the backend configuration and appears in the dropdown instantly. On subsequent visits to the Upload Page (or Library Page), this new subcategory will already be available.

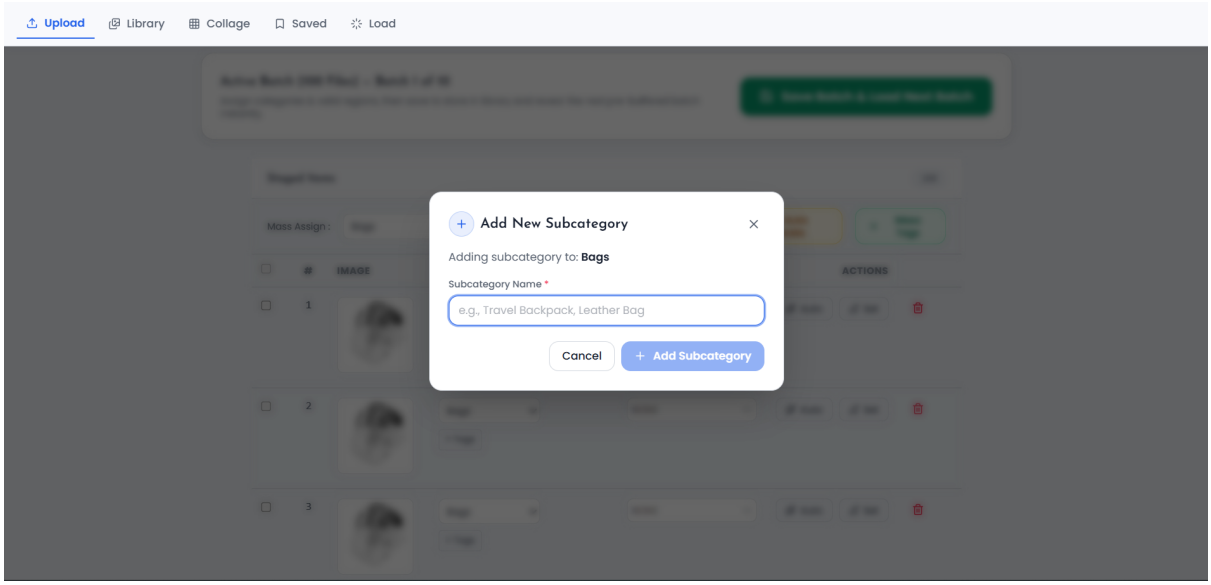


Figure 6: Add New Subcategory Box

2.3 Mass Actions and Individual Item Controls

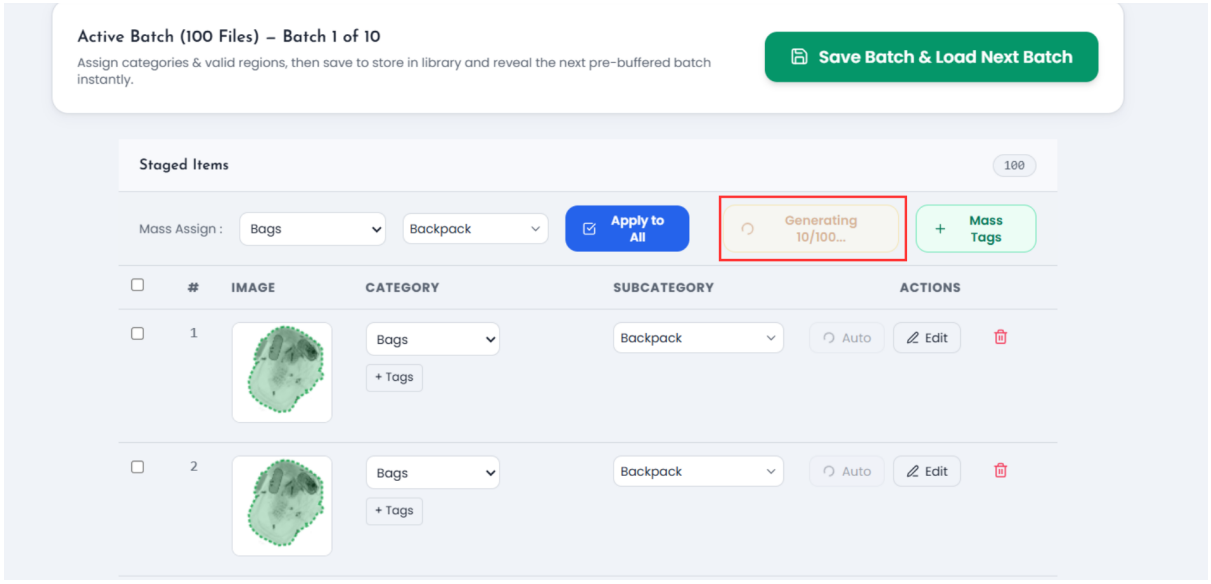


Figure 7: Mass Auto Generate Valid Regions

The **Mass Auto Generate** button triggers valid region generation for all items lacking one. Each row in the staged items table has individual controls:

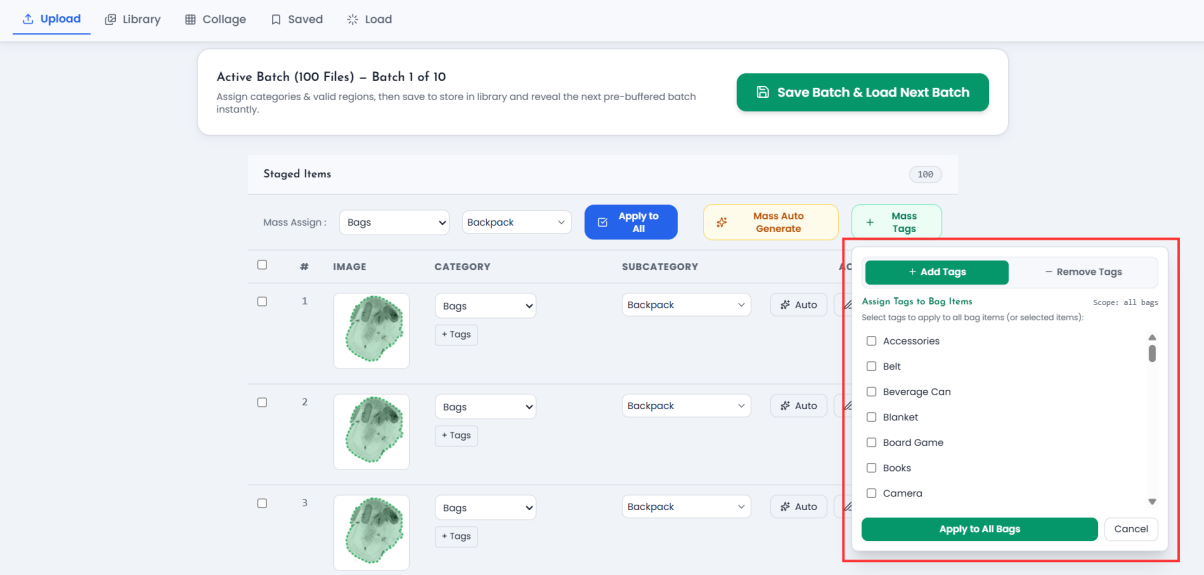


Figure 8: Mass Assign Tags

The **Mass Assign Tags** button opens a tag selector panel where the user can check tags to add to all selected items. The panel includes a search box to find tags quickly.

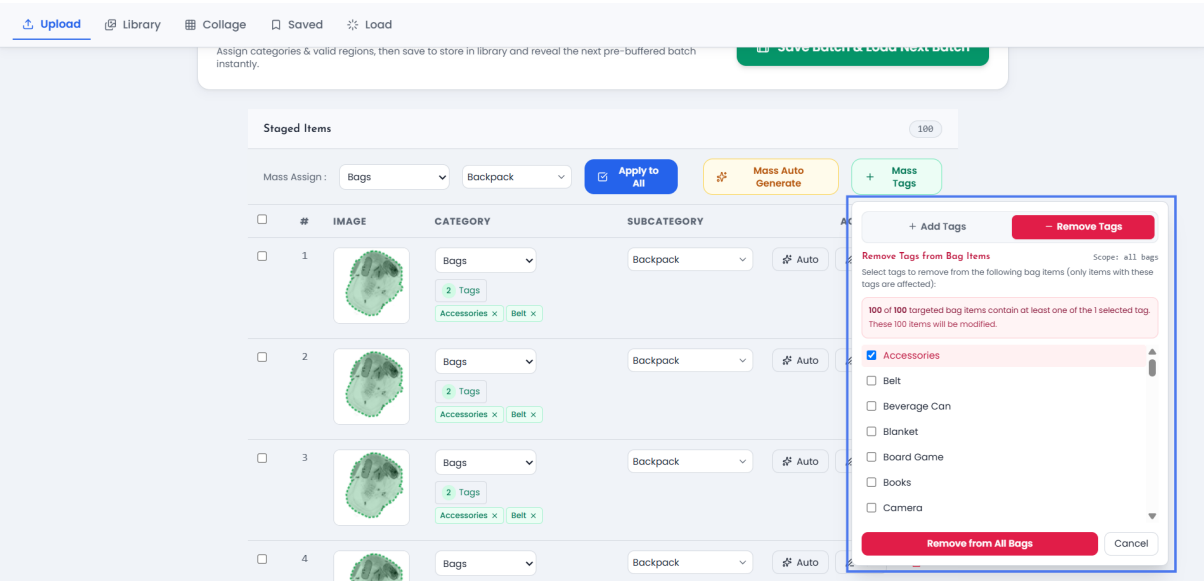


Figure 9: Mass Tags Assigned

After applying tags, the assigned tag labels are visible directly in the table. The mass tags button also now offers a **Mass Remove Tags** option, which opens the same selector but in remove mode, allowing the user to bulk-remove tags from selected items.

2.4 Valid Region Editor

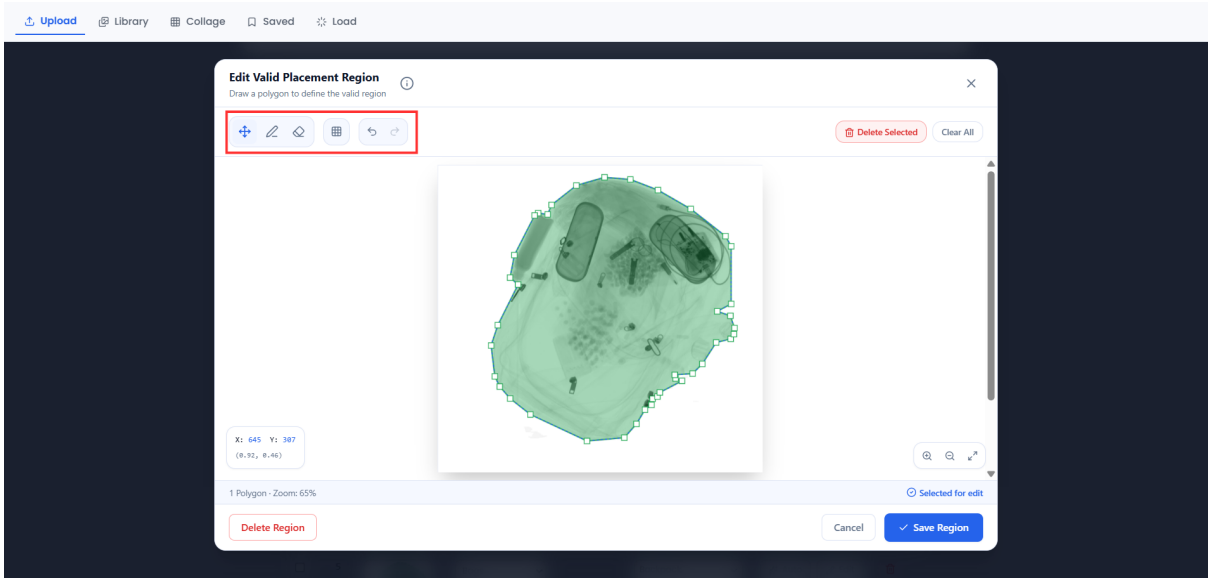


Figure 10: Singular Valid Region Editor

The valid region editor opens as a full-screen modal with the following toolbar buttons:

- **Pencil button** — Draw or edit polygon points by clicking on the image
- **Eraser button** — Remove existing polygon points by clicking on them
- **Grid button** — Toggle a grid overlay to assist with precise drawing
- **Undo/Redo** — Revert or re-apply edits
- **Save/Cancel** — Save the valid region polygon or discard changes

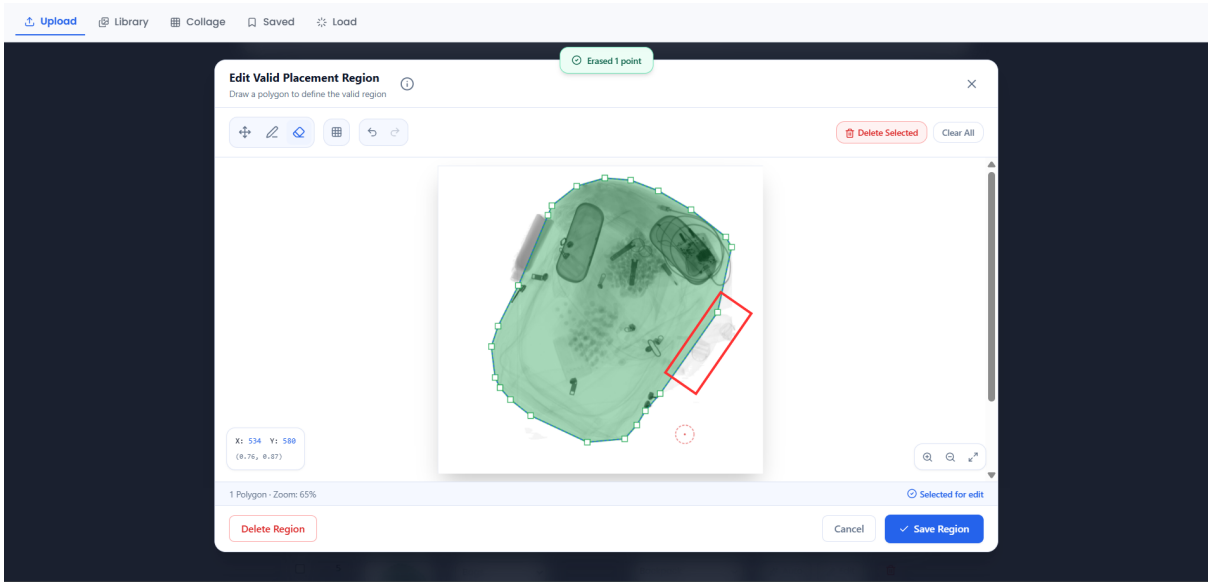


Figure 11: Valid Region Edited - Straps Removed

Users can manually refine the valid region polygon. For example, the auto-generated region may include the bag straps, which are not part of the bag's contents. Using the eraser tool, the user

removes those points from the polygon, tightening the valid region to the actual bag boundary. The edited region is saved as updated JSON polygon points.

2.5 Saving and Edge Cases

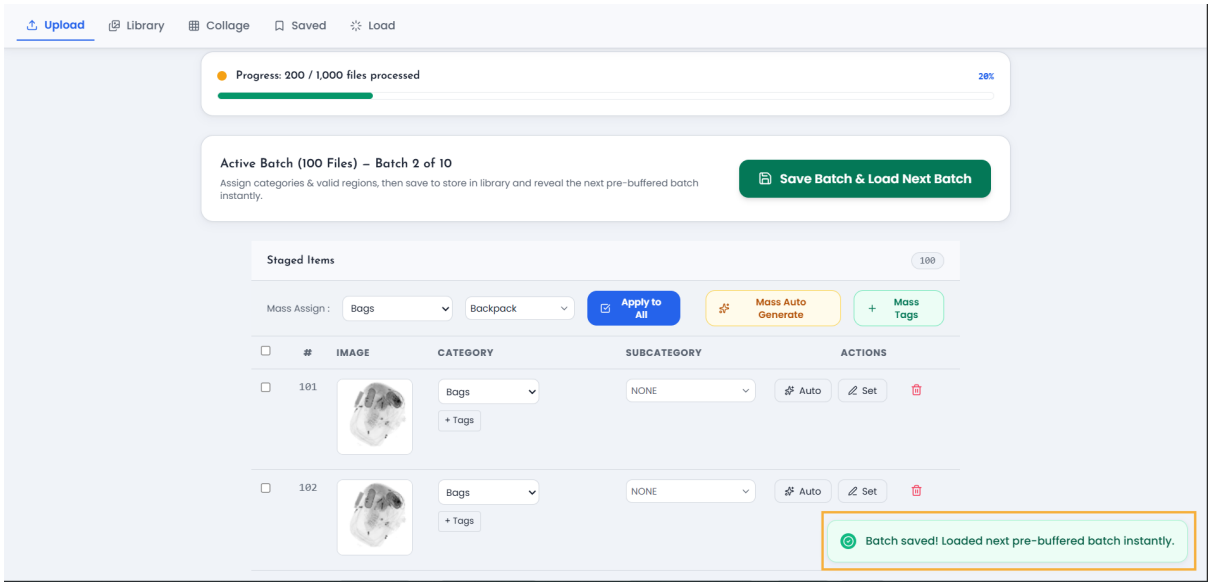


Figure 12: Save Batch and Load Next

When the user clicks **Save Batch**, the 100 staged items are persisted to the library (database + pixel data). Immediately after, the next 100 items from the buffer are loaded into the table, so the user can continue assigning categories and generating valid regions without interruption.

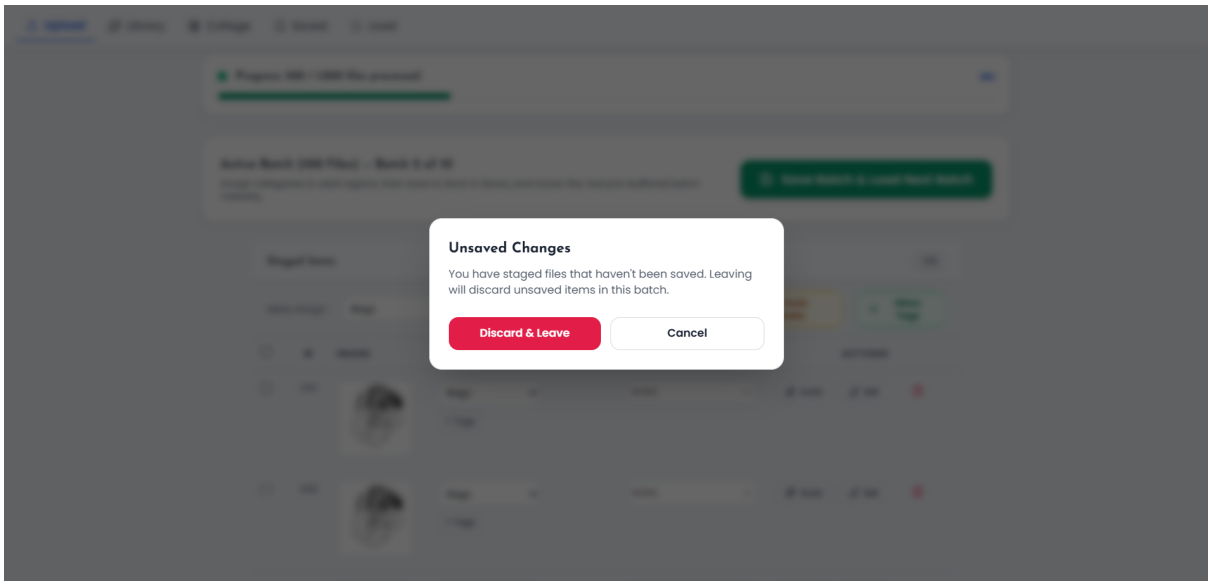


Figure 13: Unsaved Changes Popup

When the user attempts to navigate away or reload the page with unsaved changes (items that have been staged but not saved), a browser confirmation popup appears: **"Unsaved changes are there. You can discard them or cancel it."** This prevents accidental data loss.

When uploading files that already exist in the library, the system detects duplicates by filename and skips them. For example, if the user uploads 1000 files and 100 of them are already present,

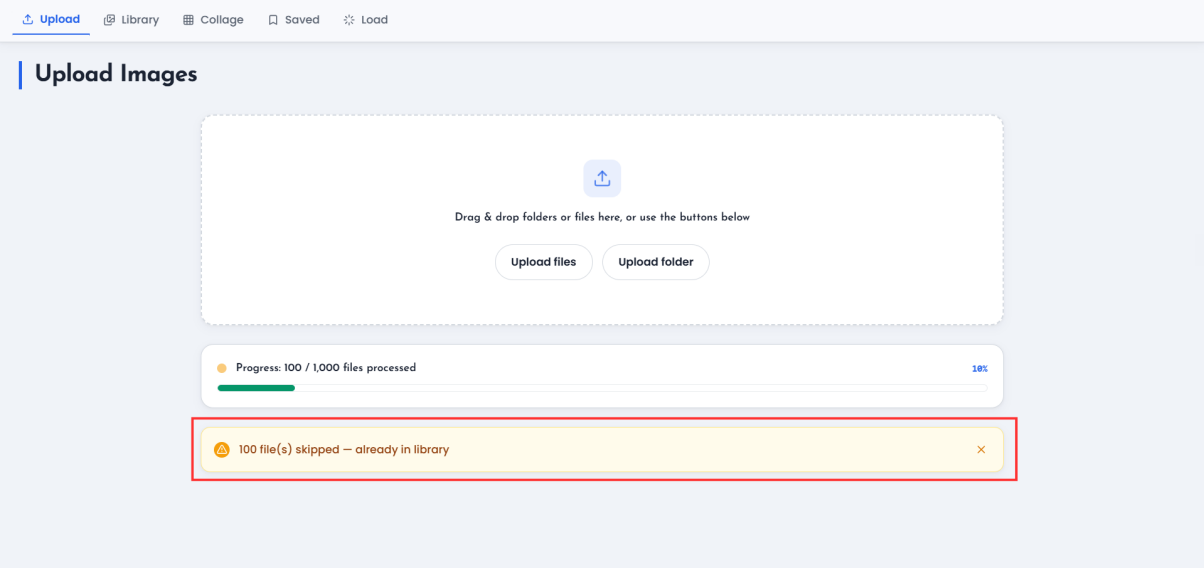


Figure 14: Duplicate Skip

those 100 are marked as skipped with the reason **"already exists"**, while the remaining 900 are processed normally. This prevents re-creating existing items.

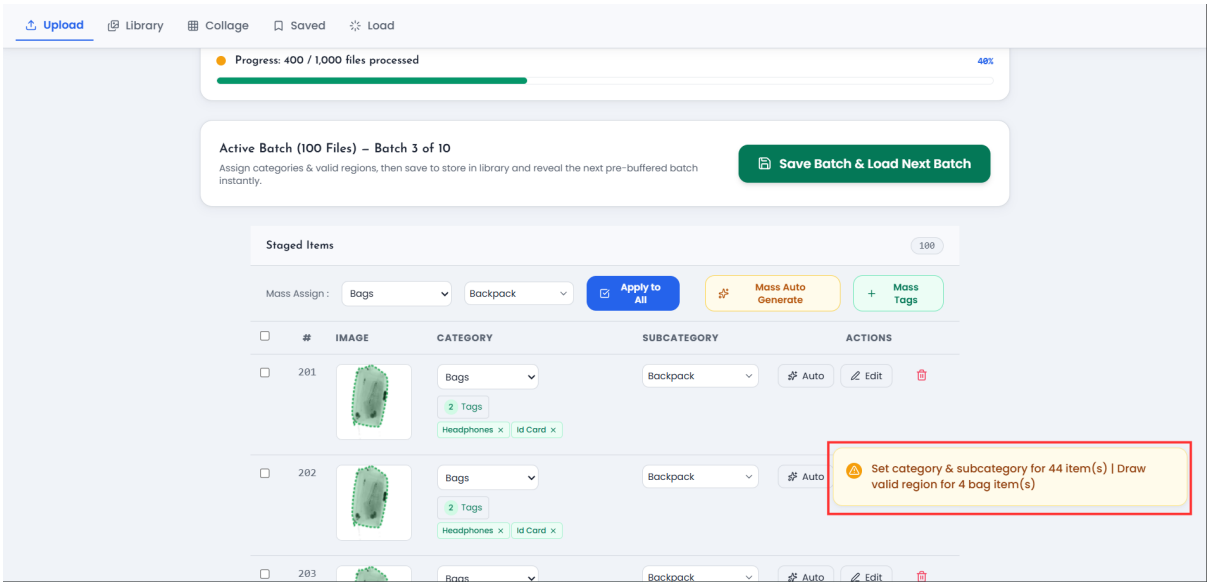


Figure 15: Save Without Valid Region Error

If the user attempts to save a batch without assigning valid regions to items that require them, an error is shown. The user must generate or manually assign valid regions for all items before the save operation will complete.

2.6 Key APIs

- `POST /api/process-raw-files` — Process raw scanner files with worker pool
- `POST /api/items/batch-save` — Persist staged items to library
- `POST /api/auto-valid-region` — Auto-generate valid region (single item)

- `POST /api/auto-valid-region-batch` — Batch auto-generate valid regions
- `PUT /api/items/:id/valid-region` — Update valid region for item
- `DELETE /api/items/:id/valid-region` — Delete valid region for item
- `GET /api/system/info` — Get system hardware info and worker count
- `POST /api/system/memory-check` — Pre-upload memory availability check
- `POST /api/temp-files/batch-delete` — Cleanup temp files for staged items
- `GET /api/items/categorized` — Fetch items for upload staging

3 Library Page

3.1 Page Layout

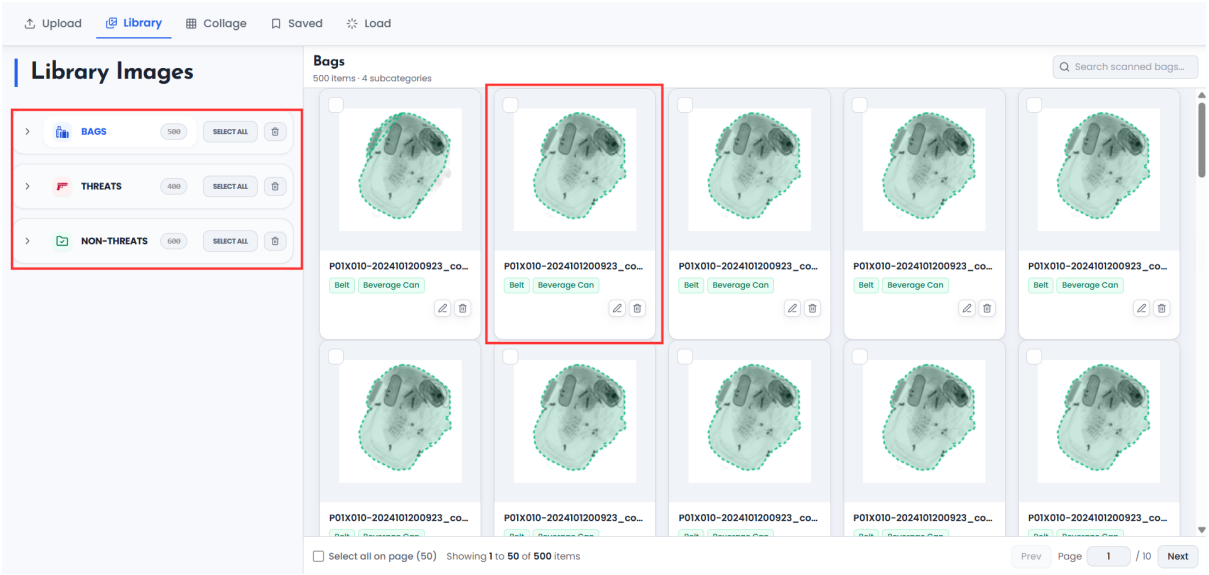


Figure 16: Library Page with Sidebar and Items

When the user navigates to the Library page, the left sidebar displays three category cards: **bags**, **threats**, and **non-threats**. Each card shows a count of items. The right side displays a virtualized grid of item thumbnails.

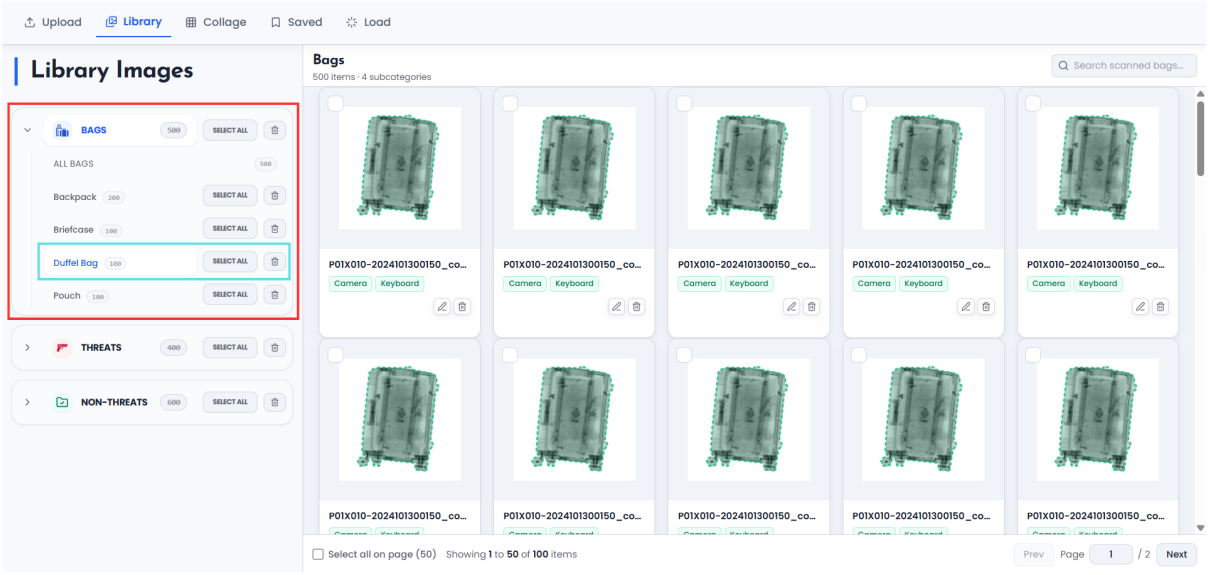


Figure 17: Bags Subcategory Selection

Expanding the **bags** category reveals its subcategories in a dropdown. Each subcategory shows the count of items it contains. Selecting a subcategory updates the main content area to show only items belonging to that subcategory.

When the user expands **threats** and clicks a subcategory, the right panel updates to show only the items under that subcategory. The same behavior applies to **non-threats and bags**.

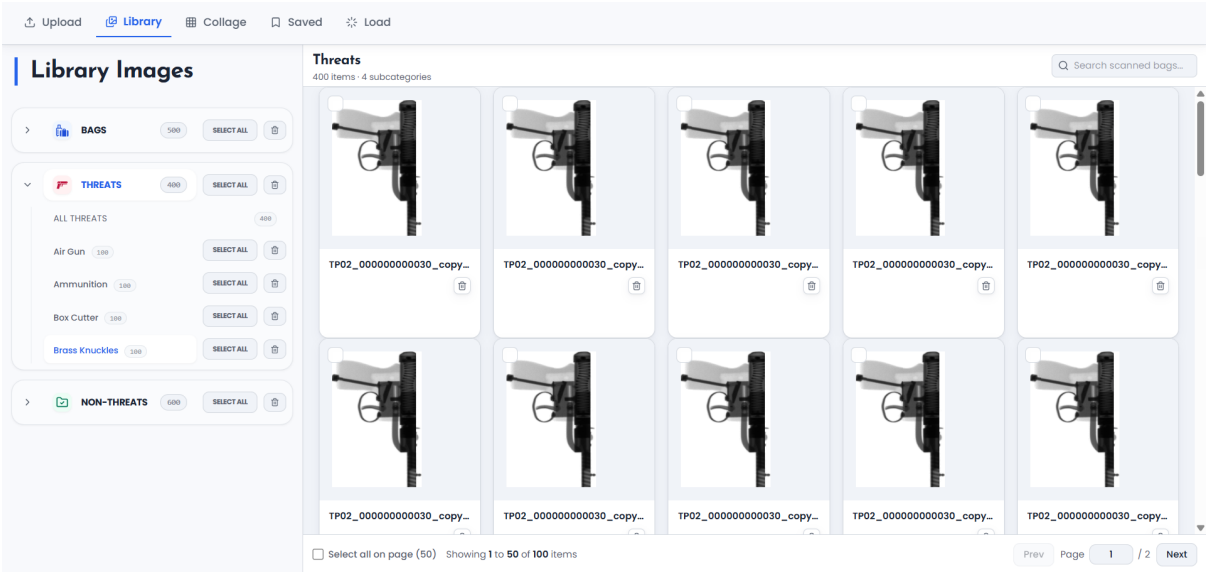


Figure 18: Threats Subcategory Selection

3.2 Selection and Generation Controls

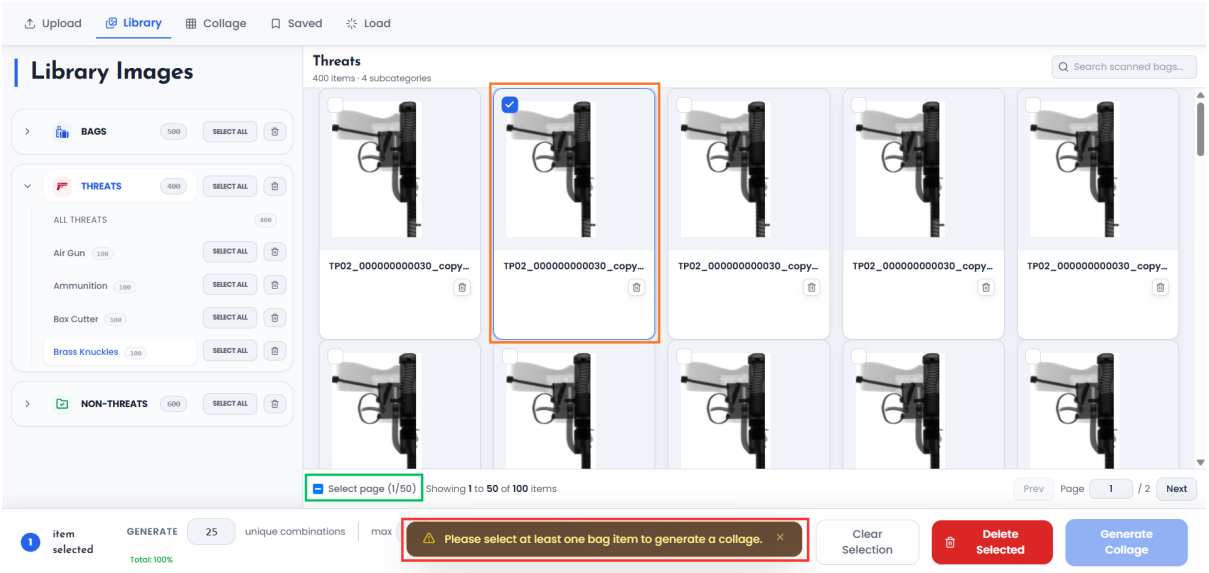


Figure 19: Select Single Item with Popup

When the user selects a single item (or items from only one category), a validation popup appears:

- **Selecting only threats:** "Select at least one bag to generate combinations."
- **Selecting only bags:** "Select at least one threat to generate combinations."
- **Selecting only non-threats:** "Select at least one bag to generate combinations."

This ensures the user has the minimum required items (one bag, at least one threat) to generate a valid Syntheic X-ray combination.

The user selects all items across categories:

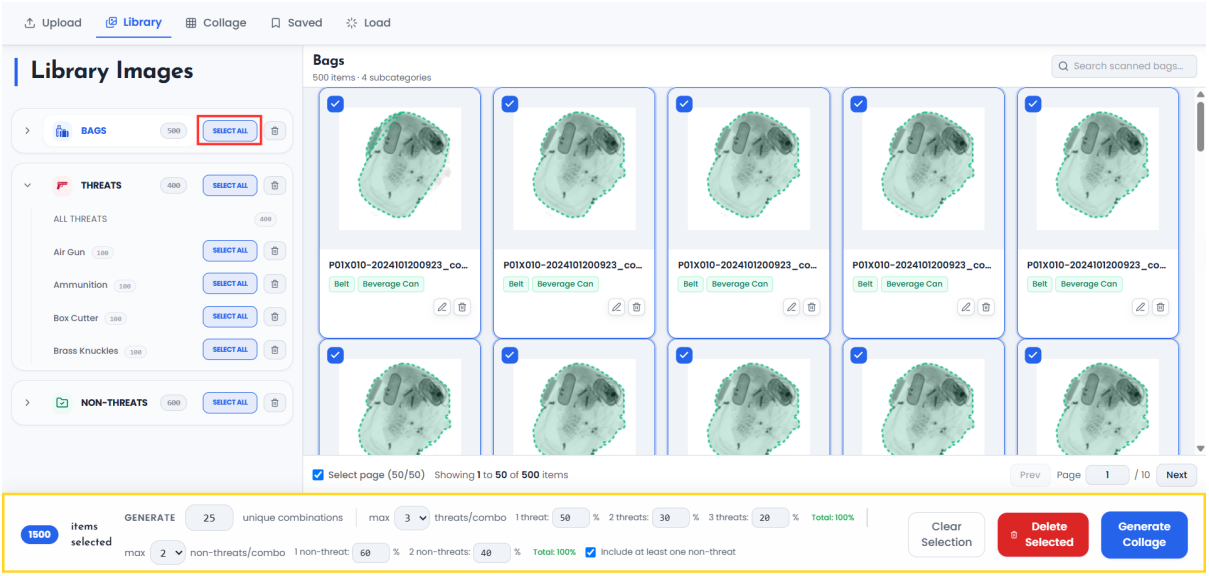


Figure 20: Select All Items and Toolbar

1. **Select all bags** — Click "Select All" on the bags category card
2. **Select all threats** — Click "Select All" on the threats category card
3. **Select all non-threats** — Click "Select All" on the non-threats category card

The selection tray, highlighted with yellow,which is also the heart of the whole project,at the bottom of the page shows:

- **Selected count** — Total items selected across all categories
- **Threat count** — Number of threats available in selection
- **K-value** — Slider for number of combinations to generate
- **Max threats** — Selector for maximum threats per image (1, 2, 3, 4)
- **Max Non-threats** — Selector for maximum non-threats per image (1, 2, 3, 4)
- **Distribution percentages** — Percentage controls for threats and non-threats in each combination
- **Generate Collage** — Button to include atleast one non-threat per combination
- **Include at least 1 Non-threat** — Button to navigate to the Collage page with selected items

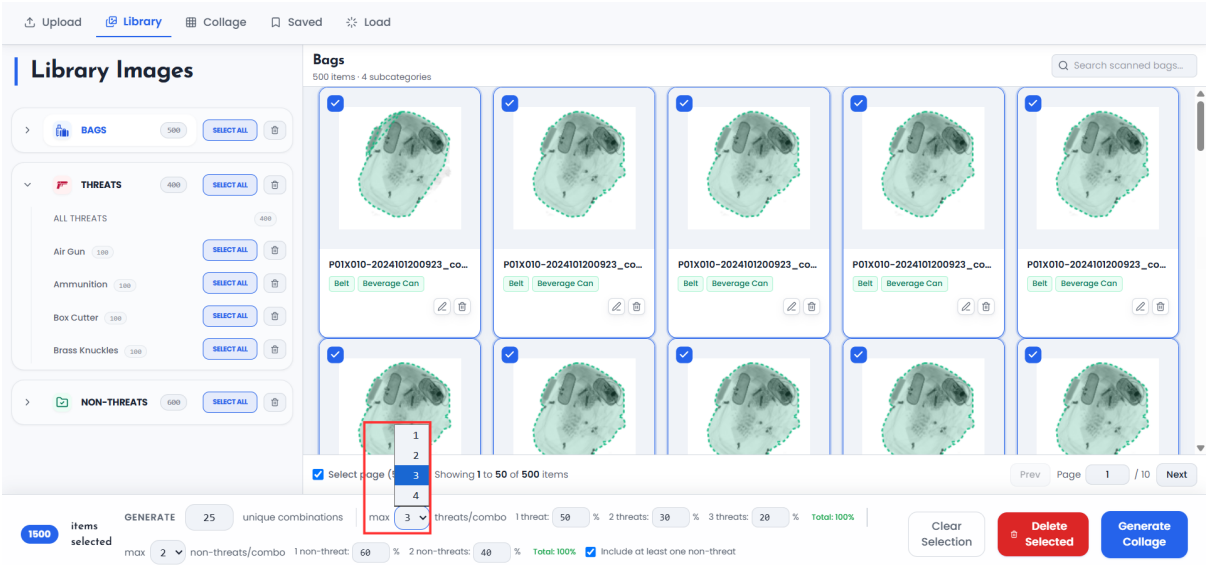


Figure 21: Max Threats and Distribution

Using the generation toolbar:

- **Max Threats:** Set to 1, 2, 3, or 4 to control how many threats appear per X-ray image
- **Distribution percentages:** Allocate percentages between threats and non-threats — the total must equal 100%. The same slider controls exist for non-threats.

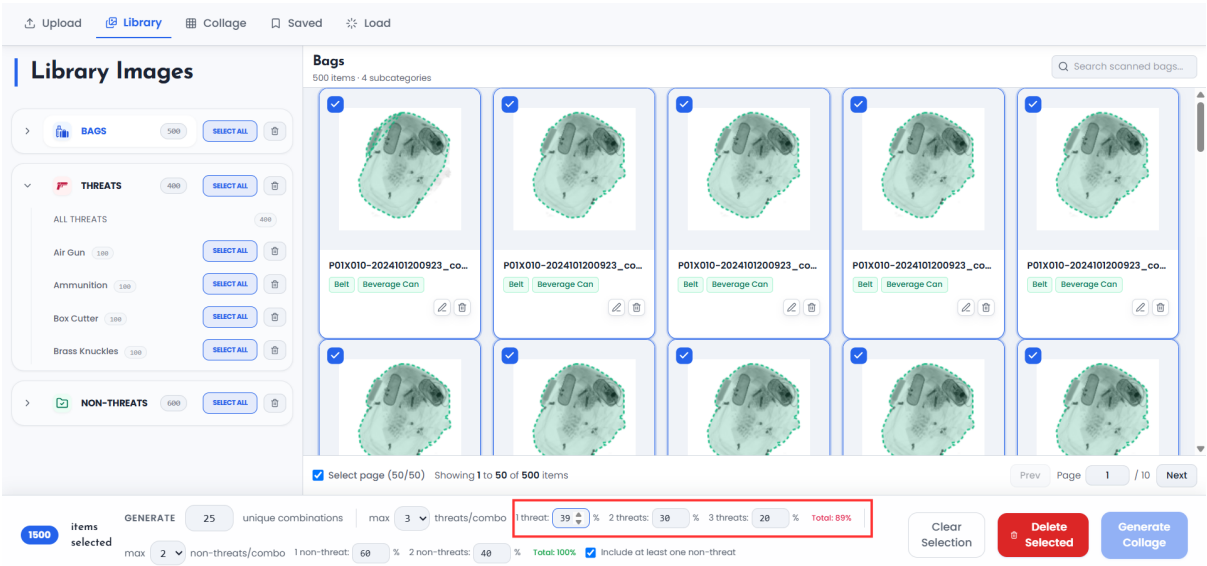


Figure 22: Percentage Error and K-Value Notification

When the percentages do not add up to 100%:

- The **Generate** button is disabled
- The **remaining percentage** is shown (e.g., “Total 89% ” in red)
- The user must adjust the sliders to make them total exactly 100%

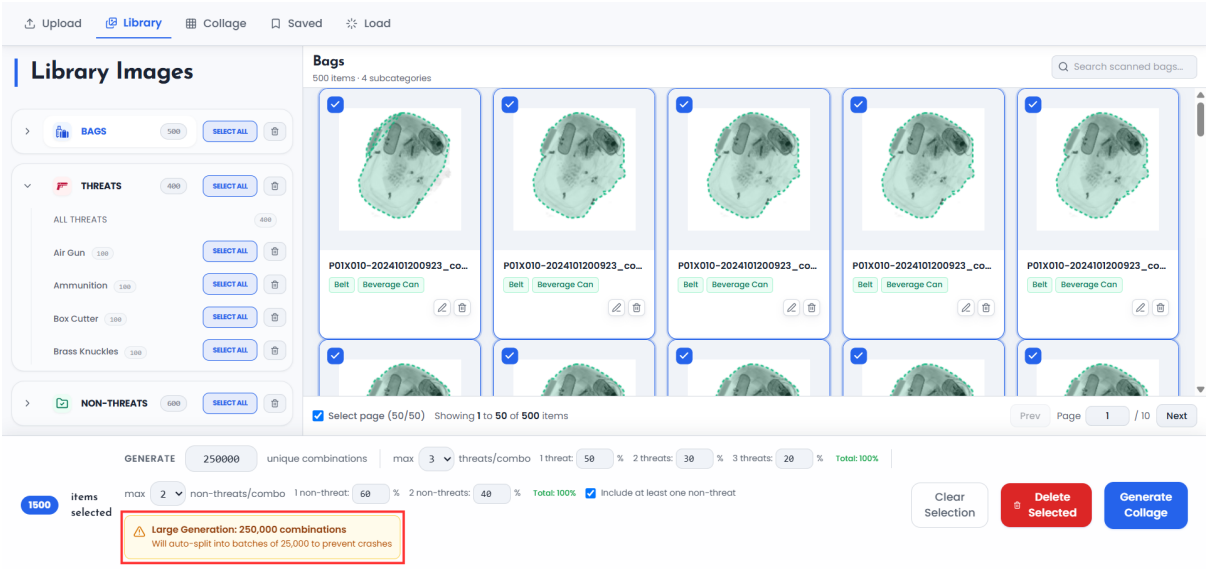


Figure 23: Large number handling

Additionally, when the K-value is set to a large number (e.g., 250,000), the system automatically shows a notification that the combinations will be split into batches of 25,000 to prevent memory crashes and ensure stable generation.

3.3 Key APIs

- `GET /api/items/categorized` — Fetch all items organized by category/subcategory
- `GET /api/items/:id` — Fetch single item details
- `GET /api/items/:id/pixel-data` — Retrieve pixel data for rendering
- `PUT /api/items/:id` — Update item metadata (name, category, tags)
- `DELETE /api/items/:id` — Delete item and its pixel data
- `POST /api/items/batch-save` — Batch persist items from upload
- `POST /api/items/bulk-delete` — Bulk delete multiple items
- `GET /api/custom-subcategories` — Fetch custom subcategories
- `POST /api/custom-subcategories` — Add new custom subcategory
- `DELETE /api/custom-subcategories/:category/:slug` — Remove custom subcategory
- `GET /api/storage/catalog` — Get storage catalog for diagnostics

3.4 Item Editing and Deletion

Item editing supports renaming files, changing category/subcategory.

Deletion behavior:

- **Single item:** Removes from library and deletes pixel data file
- **Category:** Removes all items in category and their pixel data
- **Confirmation dialog** required for both

4 Collage Page

4.1 Page Layout

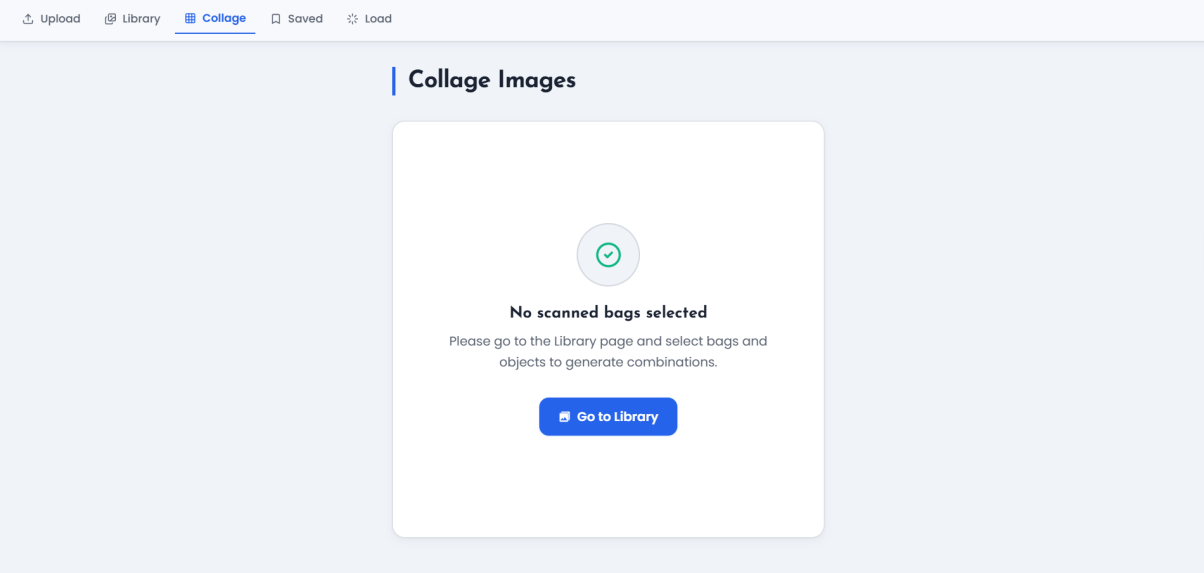


Figure 24: Collage Page Empty State

When no combinations have been generated yet, the Collage Page shows an empty state with the message **"No scanned bags selected"** and a **"Go to Library"** button to navigate back and select items.

4.2 Generation and Initial View

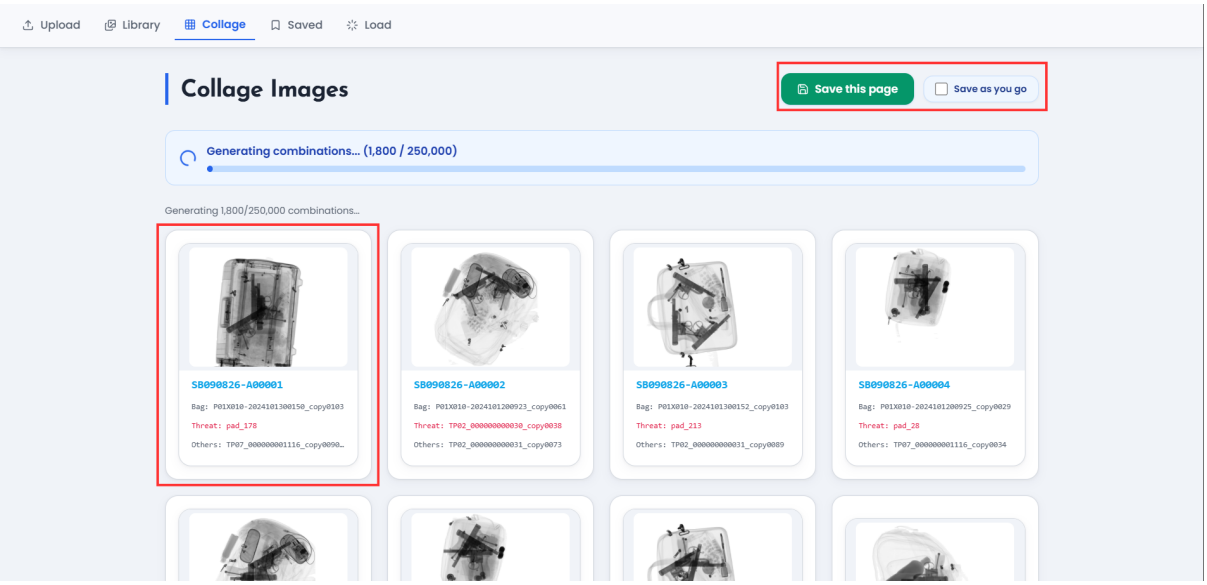


Figure 25: Generate Combinations from Library

After selecting items in the Library page and clicking **Generate Combinations**, the user is redirected to the Collage page. A progress banner appears showing generation progress (e.g., "1,800 / 250,000"), and the first 100 combinations are displayed in the grid. The **Save this page** and **Save as you go** button are visible in the header. The item cards show the preview image of the combinations and the list of objetcs in it.

4.3 Combination Preview

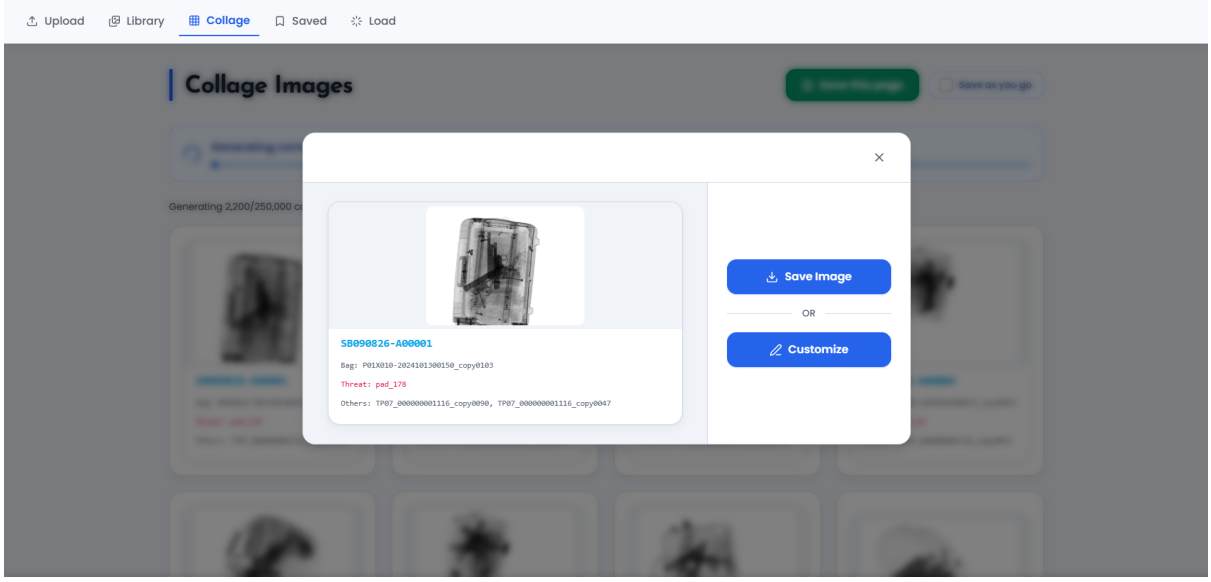


Figure 26: Combination Card Mini Preview

When the user clicks on any combination card, a mini preview box opens showing:

- **Synthesized bag preview** — The generated X-ray image
- **Items in combination** — List of bag, threats, and non-threats
- **Customize button** — Opens the customization page
- **Download button** — Downloads the current combination image

4.4 Customization Page

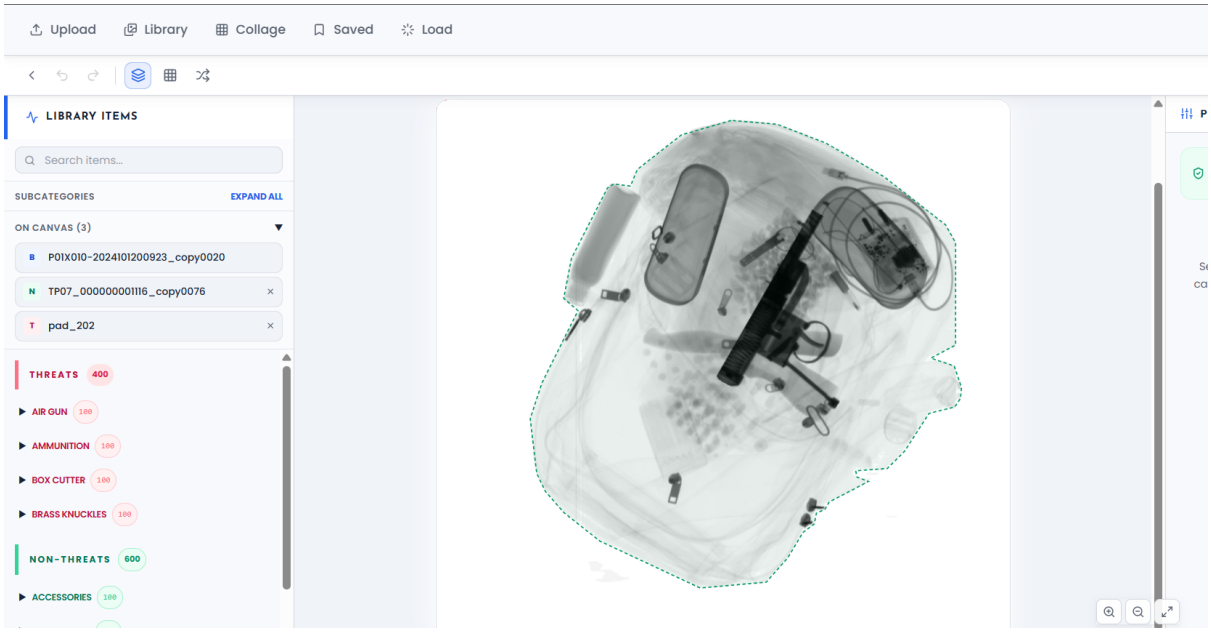


Figure 27: Customization Page

The customization page allows fine-tuning of combinations:

- **Left sidebar** — Shows objects in the current layout plus all available threats and non-threats from the library
- **Center canvas** — Displays the combination with the bag and placed objects, including the valid region overlay

4.5 Object Placement and Rotation

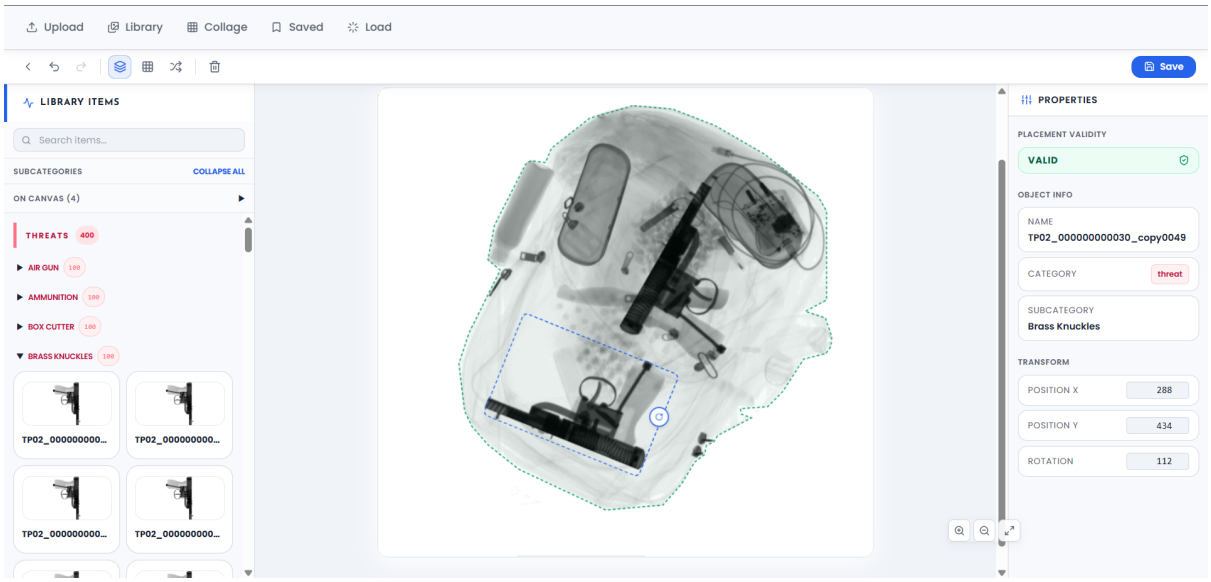


Figure 28: Add Object from Sidebar with Details

Users can add objects to the combination:

1. Click **Threats** or **Non-threats** in the left sidebar
2. Open the subcategory dropdown to see available items
3. Click an item to add it to the bag

When an object is selected:

- **Right panel** shows object details: type, position (X, Y), and rotation
- **Rotation toggle** appears — dragging the cursor rotates the object

4.6 Valid Region Validation

The system enforces valid region constraints. When an object is moved outside the bag's valid region:

- The out-of-bounds area is highlighted
- A warning appears: **"One object is out of the valid region"**
- The user must reposition the object inside the valid region before saving

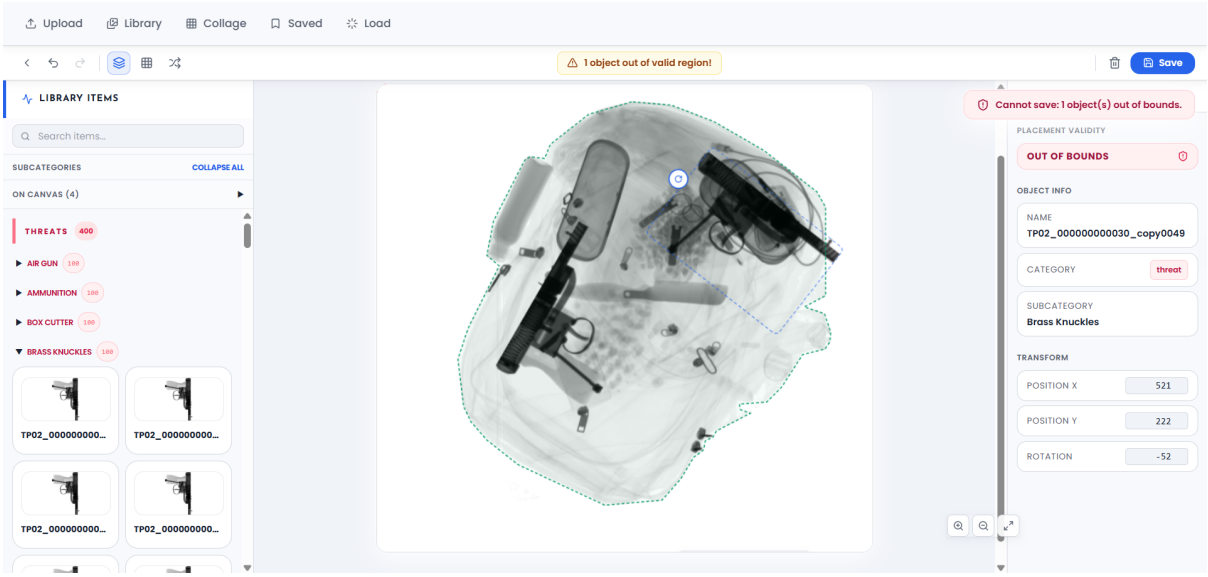


Figure 29: Object Out of Valid Region

4.7 Saving Customizations



Figure 30: Save Customization and Back

When all objects are within the valid region:

1. User clicks **Save**, highlighted in red.
2. System validates all object positions
3. Confirmation appears: **"Changes saved successfully"**
4. User clicks **Back**, highlighted in green, to return to the collage page with the updated combination

4.8 Generation Performance and Buffering

Generation performance optimizations:

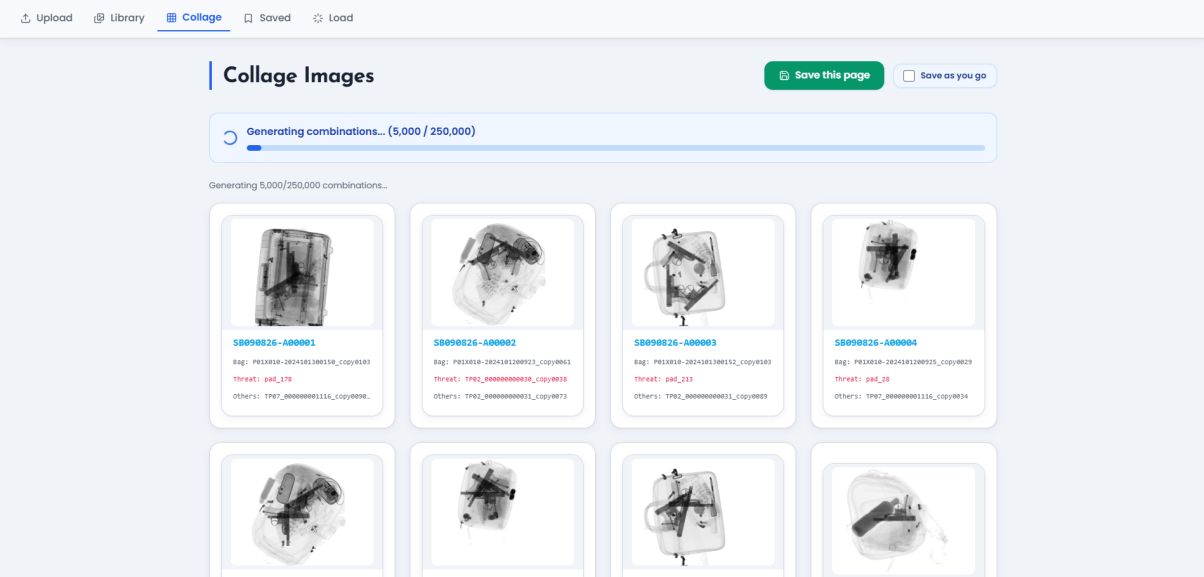


Figure 31: Generation Pausing and Buffering

- When generating large sets (e.g., 250,000 combinations), the system pauses generation periodically
- The page displays 100 combination images at a time
- 200 additional combinations are stored in an internal buffer
- At the halfway point, the next batch of 5,000 combinations is generated
- This prevents memory crashes and ensures smooth UI performance

The system can generate faster by separating rendering from saving: saving up to 2,500 combinations first, then resuming generation for the next 5,000. This avoids parallel save/generation conflicts and maintains stability.

4.9 Save This Page

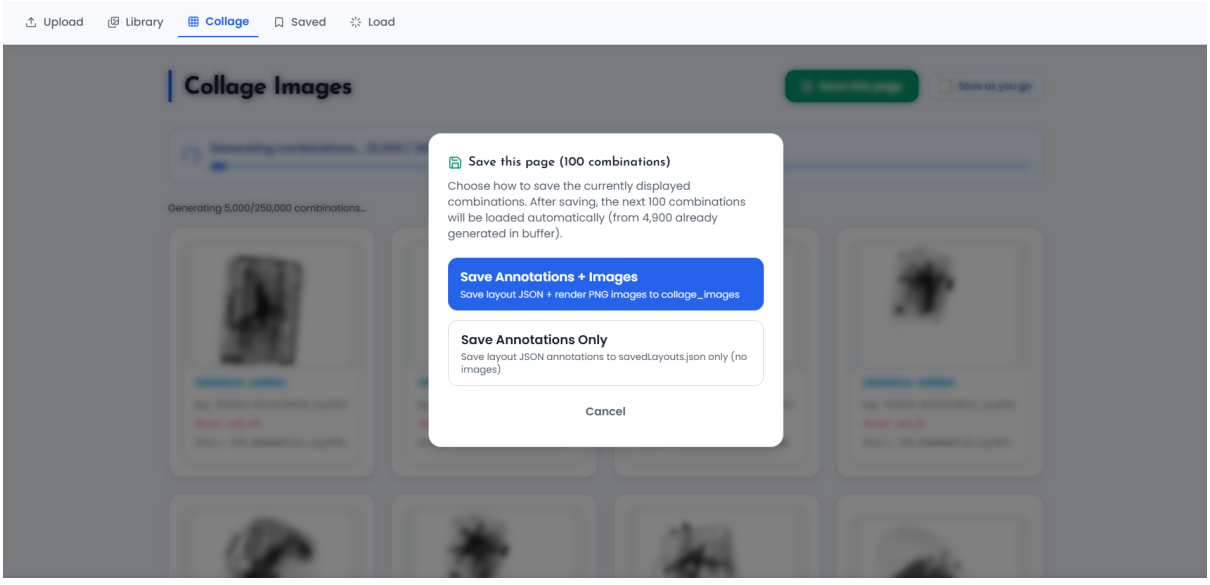


Figure 32: Save This Page Dialog

The **Save this page** button opens a dialog with options:

- **Save annotations only** — Exports only the annotation(JSON) data
- **Save annotations + Images** — Exports annotations, high and low images

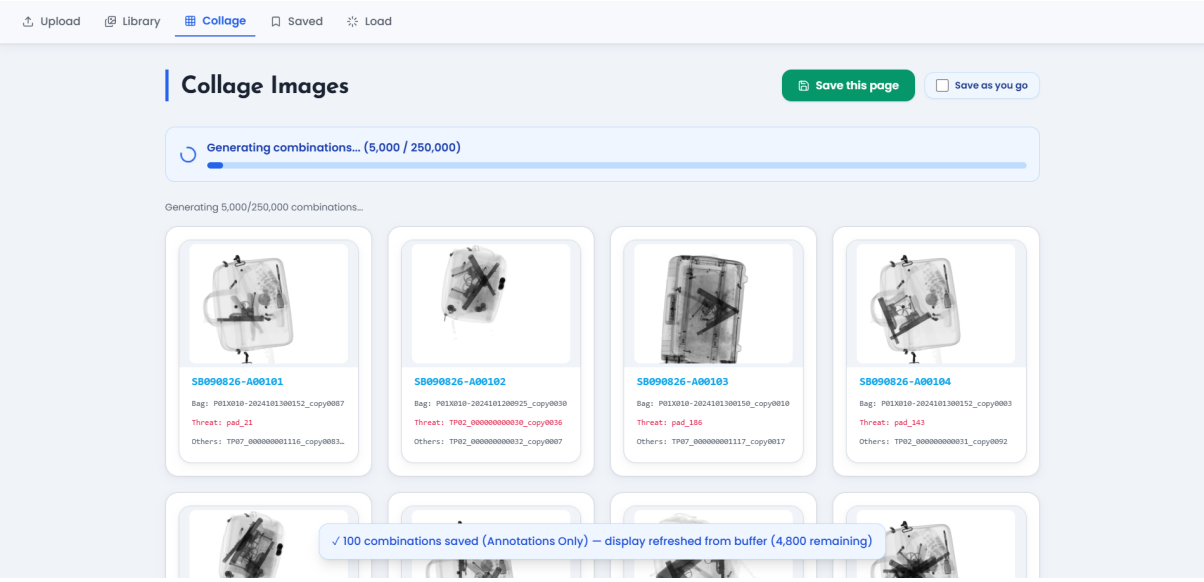


Figure 33: Save This Page Success

After saving:

- A success notification appears: **"100 combinations saved successfully"**
- The next 100 combinations are automatically loaded from the buffer into the grid
- The user can continue browsing without interruption

4.10 Save As You Go

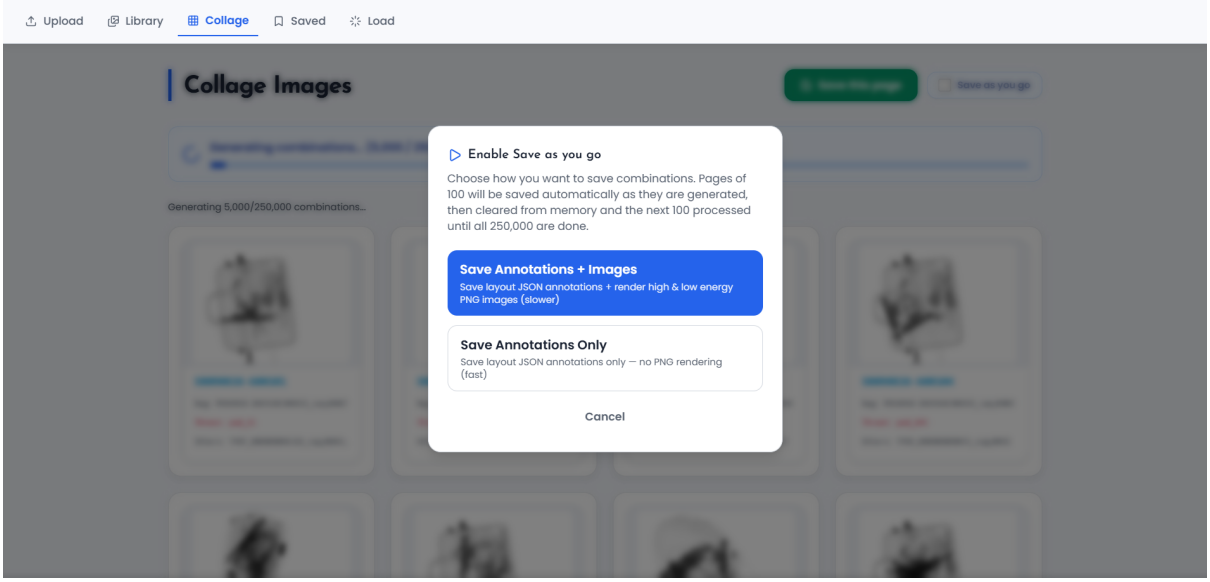


Figure 34: Save As You Go Dialog

The **Save As You Go** feature opens a dialog allowing the user to choose:

- **Only images** — Saves only the rendered PNG images
- **Only annotations** — Saves only the JSON annotation files
- **Both images and annotations** — Saves both formats

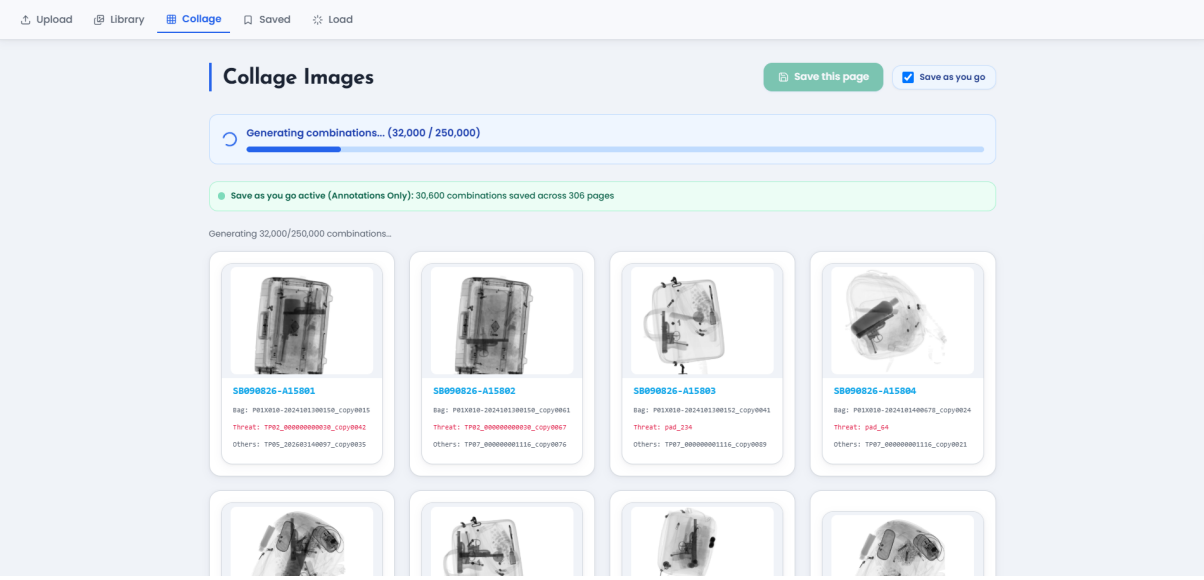


Figure 35: Save As You Go in Action

When enabled, the system automatically saves combinations in batches of 100 every 5 seconds during generation, ensuring no data loss and reducing manual work.

4.11 Navigation and Unsaved Changes

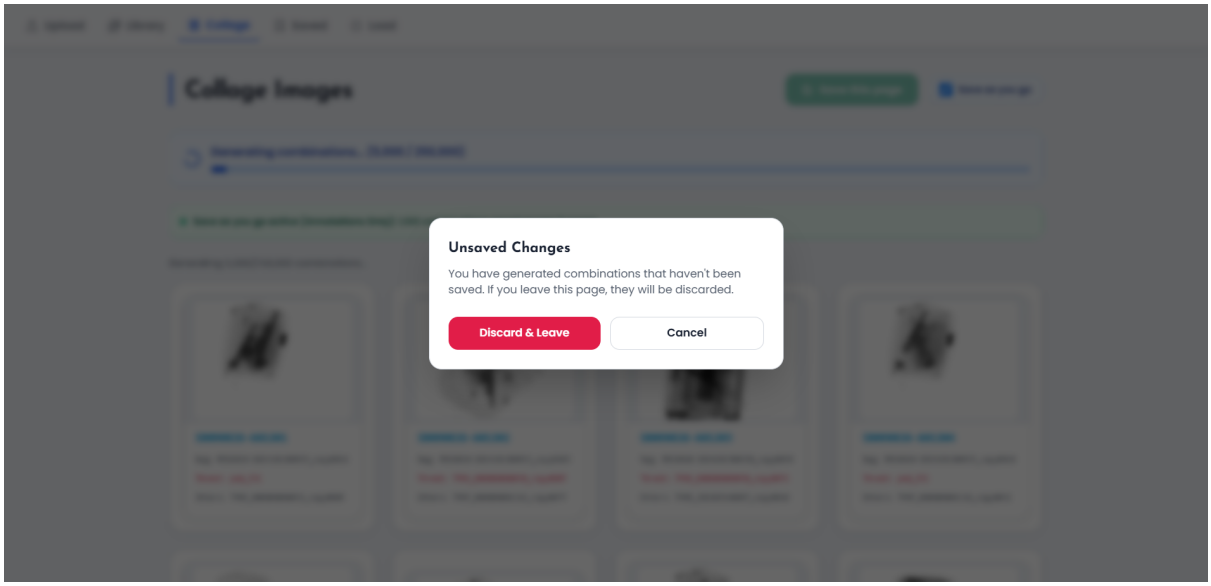


Figure 36: Unsaved Changes Dialog

When the user attempts to navigate away or reload the page with unsaved changes:

- A dialog box appears: **"Unsaved changes are there and you cannot go. You can discard the changes or cancel it."**

5 Saved Page

The Saved Page manages saved layout combinations:

- **Filter Toolbar**,marked in green — Multi-axis filtering with OR/AND logic (bag subcategory, threat subcategory, tags)
- **Layout Table** — Sortable table with selection and actions
- **Detail Modal** — Preview layout with object details
- **Export System** — Export layouts with configurable options
- **Integrity Checking** — Verify layout references and pixel data

5.1 Page Layout and Actions

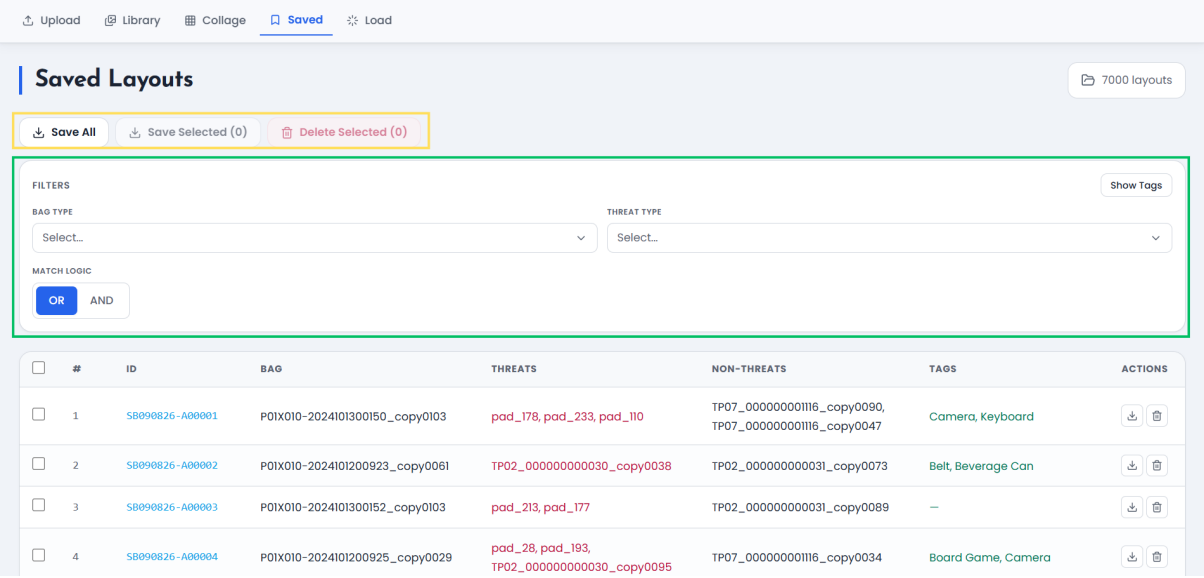


Figure 37: Saved Page Main View

When the user navigates to the Saved tab, the header displays action buttons: **Save All**, **Save Selected**, and **Delete Selected**, all highlighted in yellow. Below is the filter toolbar with bag filter dropdown, threats filter dropdown, tags filter with hide button, and OR/AND logic toggle. The layout table lists all saved combinations with columns: name, ID, bag ID, threat IDs, non-threat IDs, tags, download button, and delete button.

5.2 Detail Modal and Preview

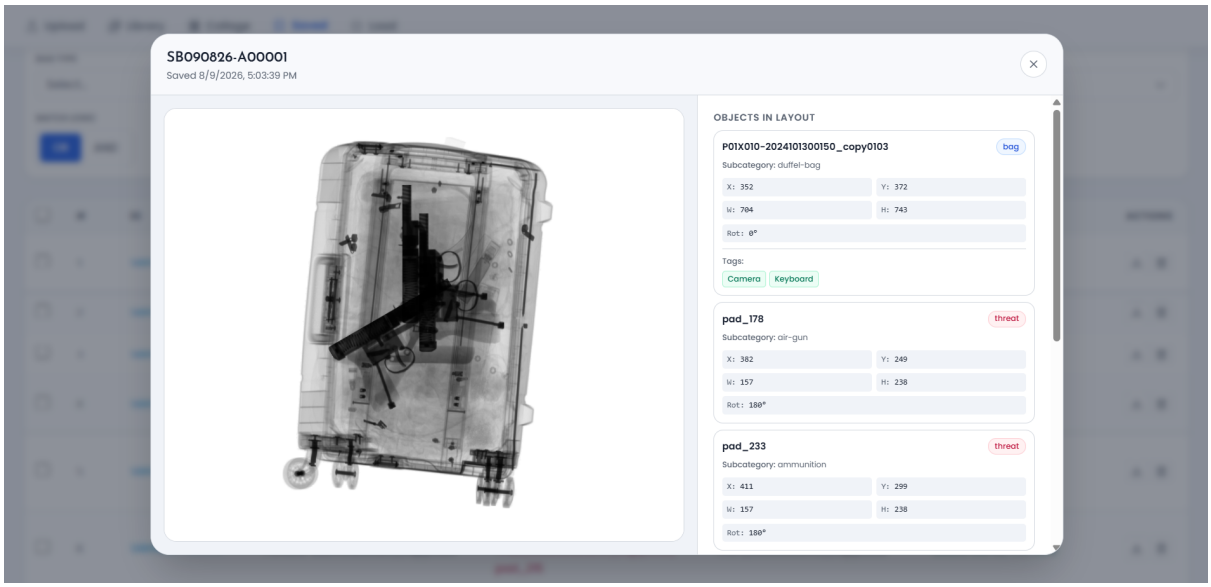


Figure 38: Saved Layout Detail Modal

Clicking any layout row opens the detail modal. The left side shows a large preview thumbnail of the generated combination displaying the bag, threats, and all present objects. The right side displays detailed information: bag name, position (X, Y coordinates), rotation angle, threat name, position, rotation, tags, and other metadata. Users can export or delete from this modal.

5.3 Filtering System

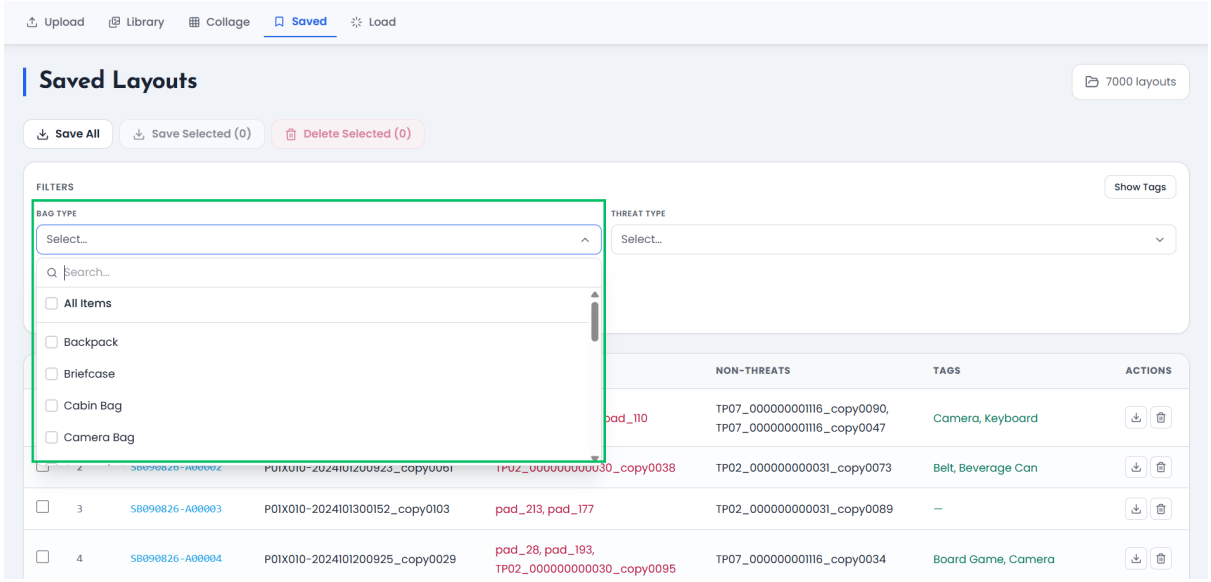


Figure 39: Bag Filter

The bag filter dropdown, marked in green, displays all available bag subcategories with a search box at the top. Users can type to search for specific bag types or scroll through the list. Selecting one or more bag types filters the layout table to show only combinations containing those bags.

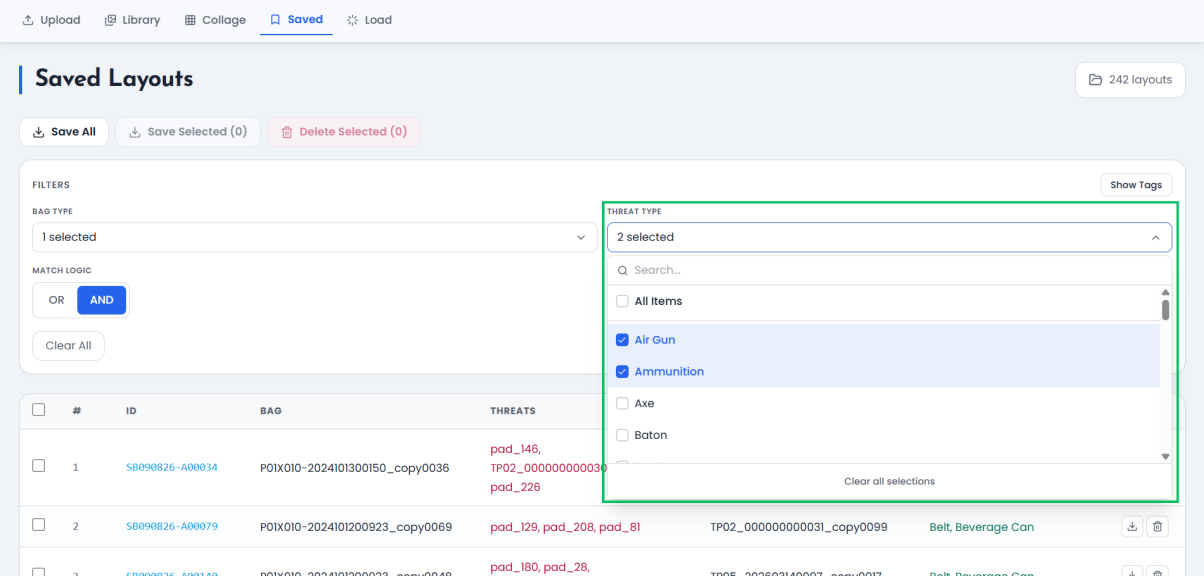


Figure 40: Threats Filter

The threats filter allows selecting multiple threat types using checkboxes. The OR/AND logic toggle determines how selections are combined: OR shows layouts with any selected threat, AND shows only layouts containing all selected threats. A search option helps find specific threats quickly.

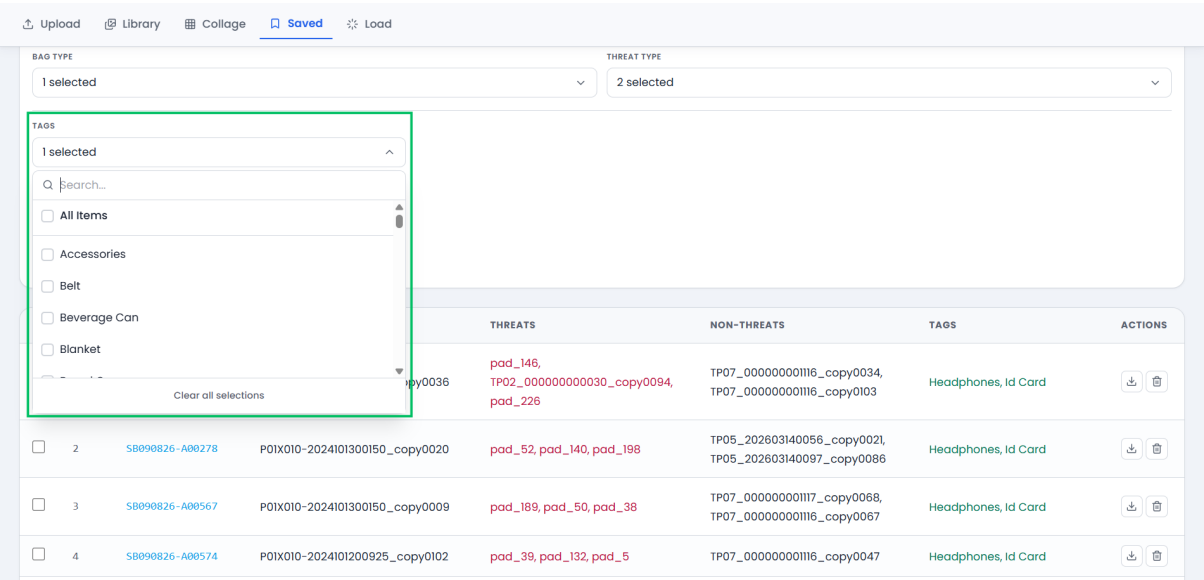


Figure 41: Tags Filter

The tags filter provides a multi-select dropdown with all available tags.It can be accessed by clicking the **Show Tags** button. Users can check multiple tags to filter layouts. The filter applies in real-time, updating the layout table as tags are selected or deselected. An "All Items" toggle allows showing all tags or only selected ones.

5.4 Export Configuration

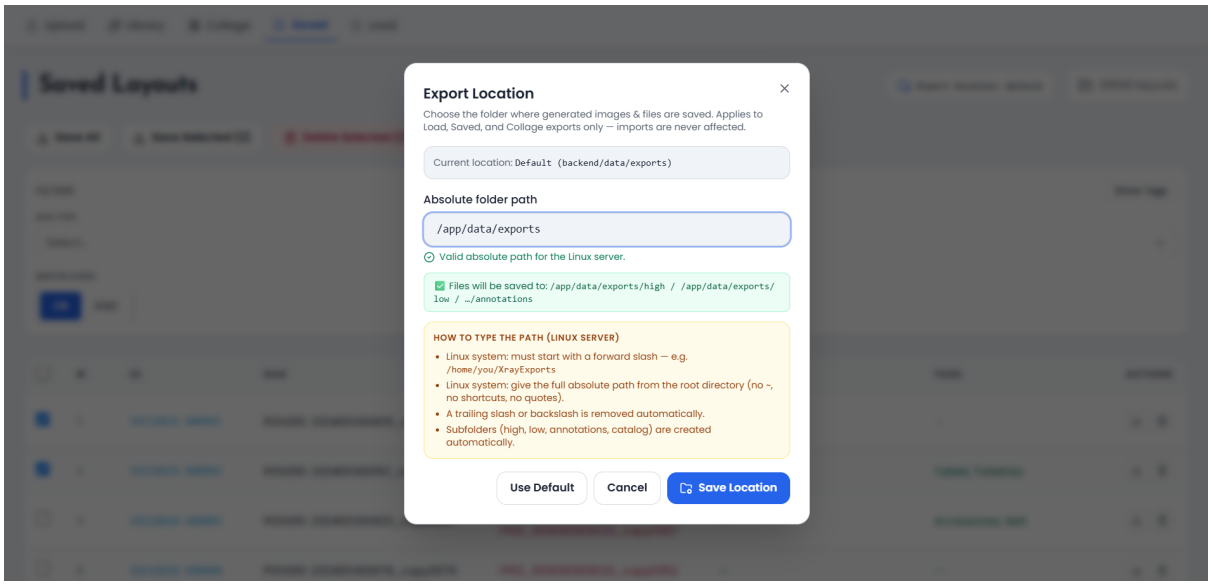


Figure 42: Exports Path Configuration

When the user first opens the Saved Page (or when configuration is needed), a modal appears asking where exports should be saved. It shows the default path option and a text input for entering a custom absolute path. The user can type the path directly.

5.5 Save Options and Notifications

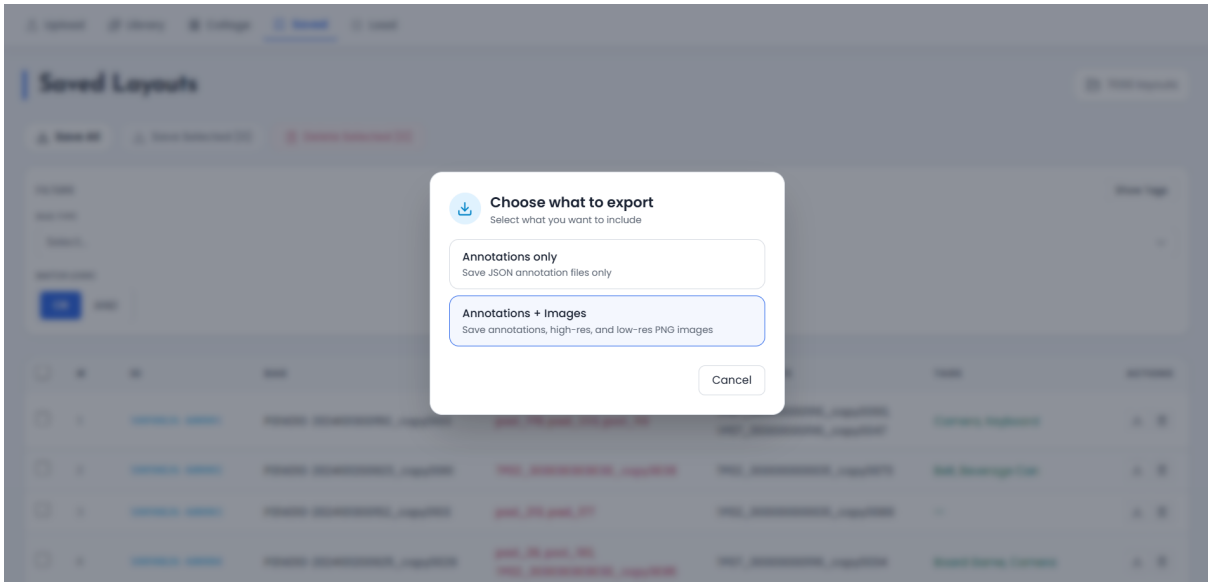


Figure 43: Save Options Dialog

When the user clicks any save button (**Save All**, **Save Selected**, or **Save Individual Files**), a popup appears asking what to save: **"Save both annotations JSON files and images (high + low resolution)"** or **"Save only annotations JSON files"**. This allows users to control the export format based on their needs.

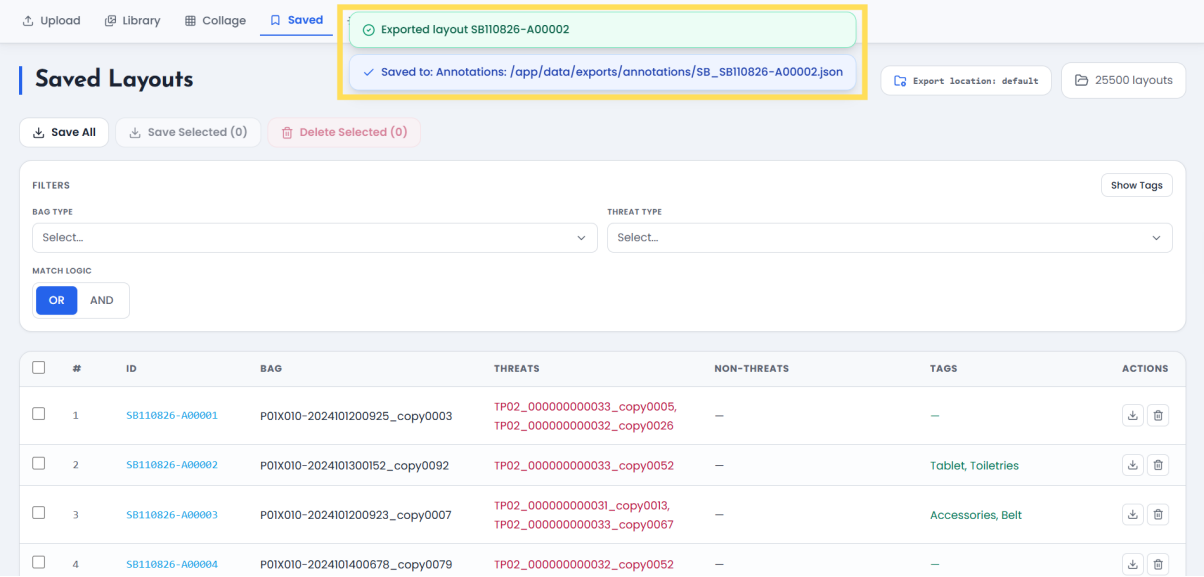


Figure 44: Individual Save Notification

After saving an individual layout, a notification appears at the top confirming the save and displaying the exact file path where the file was saved. This provides immediate feedback and helps users locate their exported files.

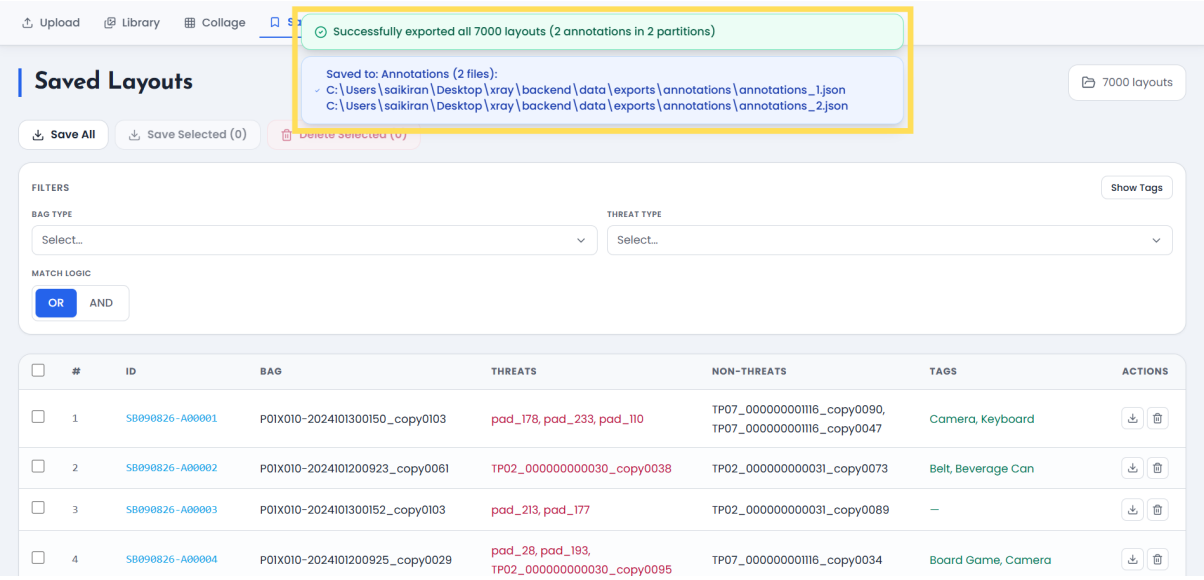


Figure 45: Save All Notification

When the user clicks **Save All** and confirms the save options, a top notification appears showing that all files were saved successfully. It displays the total number of annotation files saved and the destination path, giving users a complete summary of the export operation.

5.6 Key APIs

- `GET /api/saved-layouts` — Fetch all saved layouts
- `GET /api/saved-layouts/:id` — Fetch single layout by ID
- `POST /api/saved-layouts/batch` — Batch create layouts

- `PUT /api/saved-layouts/:id` — Update layout metadata
- `DELETE /api/saved-layouts/:id` — Delete single layout
- `POST /api/saved-layouts/batch-delete` — Bulk delete layouts
- `GET /api/saved-layouts/filtered` — Filter layouts by criteria
- `GET /api/saved-layouts/integrity` — Check layout integrity (missing items/pixel data)
- `GET /api/saved-layouts/export` — Export layouts as PNG with NDJSON streaming
- `POST /api/saved-layouts/export/cancel` — Cancel running bulk export
- `POST /api/saved-layouts/batch-bulk` — Enqueue batch for async disk write
- `GET /api/saved-layouts/batch-bulk/status` — Poll queue depth and in-flight state
- `POST /api/saved-layouts/batch-bulk/drain` — Wait for queue to drain

5.7 Integrity Checking

Integrity check verifies:

- Layout references to existing items
- Pixel data file availability
- Orphaned layout detection

5.8 Backend Storage

Backend storage:

- Layouts stored in `savedLayouts_*.json`
- Images exported to configurable folder

6 Load Page

6.1 Page Layout

The Load Page is the annotation import and rendering hub. It accepts JSON annotation files, checks every referenced object against the current library, and renders only the valid combinations as images. The page flows through four distinct phases: **Upload**, **Analyzing**, **Report**, and **Generating/Results**.

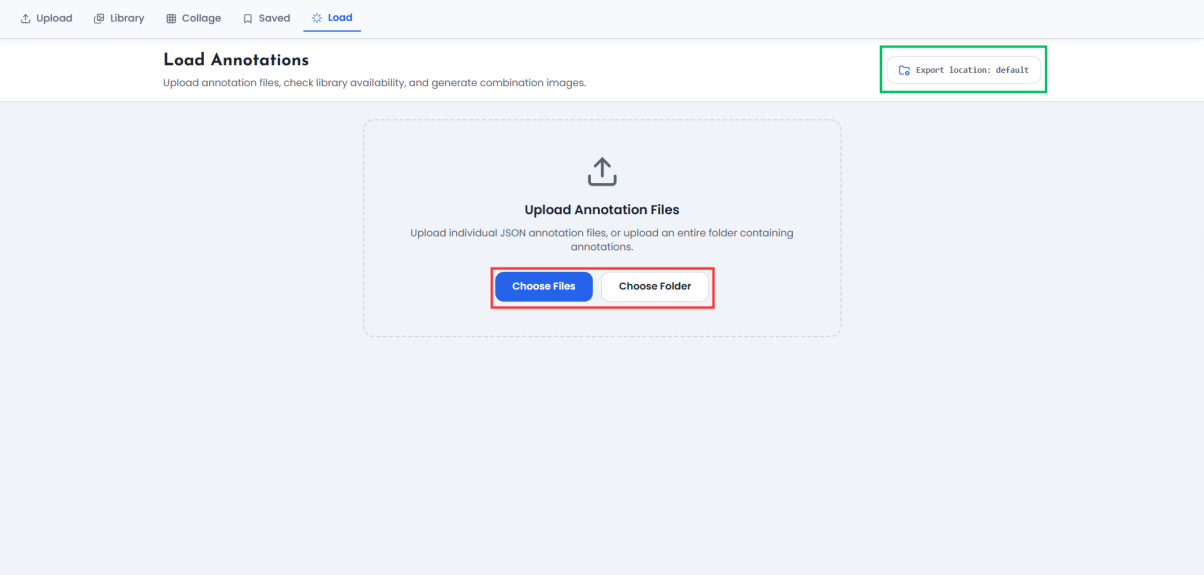


Figure 46: Load Page — Upload Zone

When the page opens, the user sees a centered drag-and-drop zone. Two primary buttons are provided:

- **Choose Files** — Opens a file picker for selecting one or more `.json` annotation files
- **Choose Folder** — Opens a folder picker that recursively scans for `.json` files using `webkitGetAsEntry`, including nested folders

Users can also drag files directly onto the drop zone. Only `.json` files are accepted; non-JSON files are ignored with an error toast. The drop zone highlights on hover.

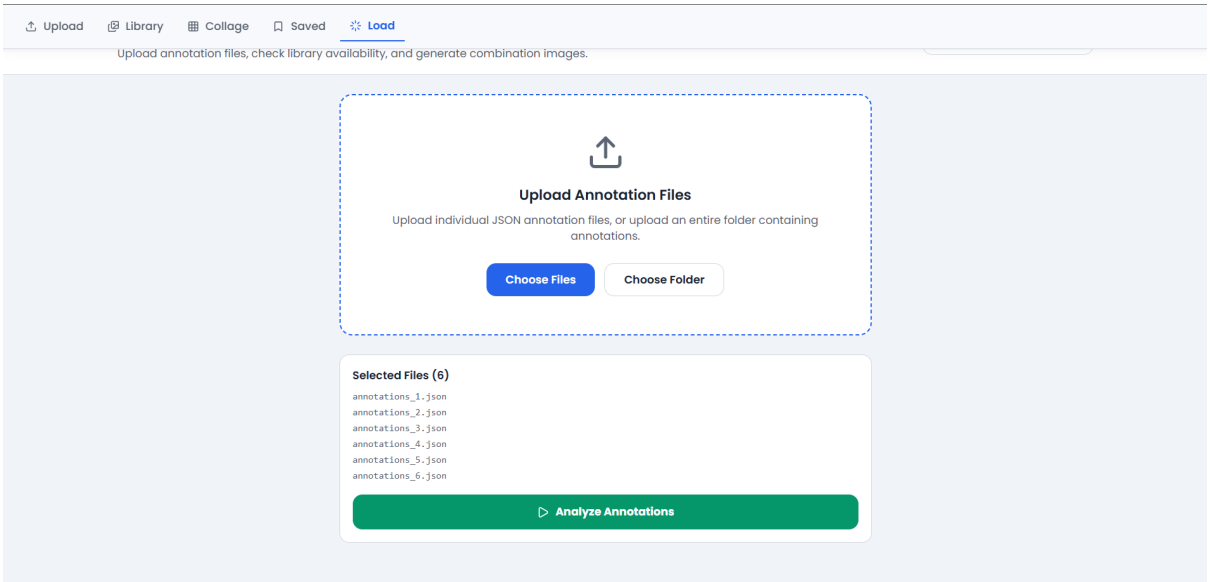


Figure 47: Selected Files and Analyze Button

After selecting files, the page lists every uploaded file by name in a scrollable panel. The **Analyze Annotations** button appears below the list. The user clicks this button to start the analysis.

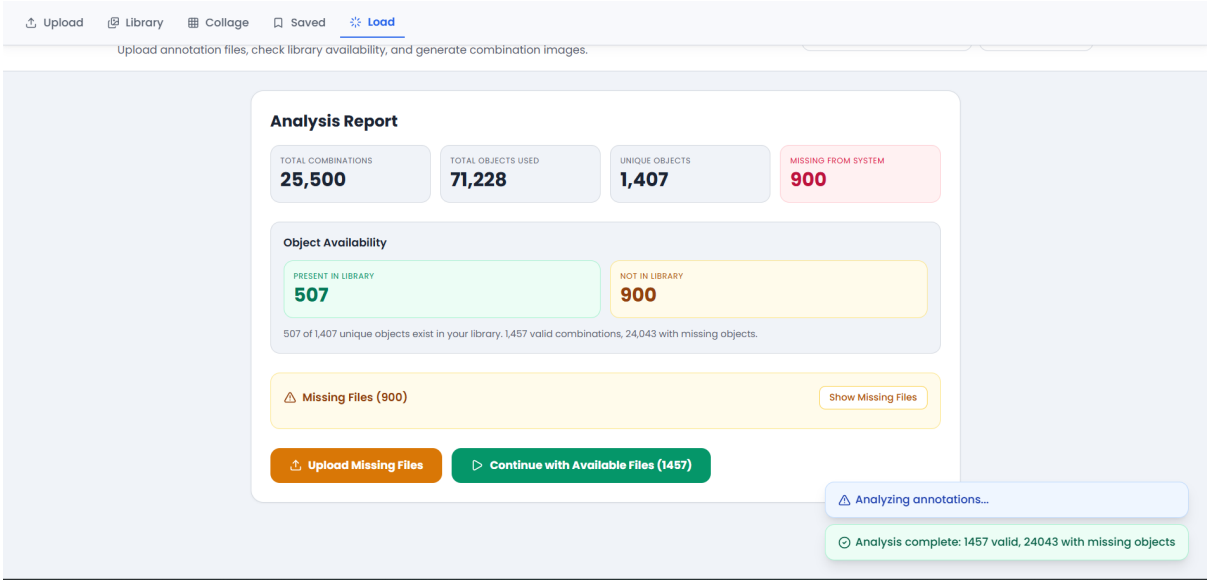


Figure 48: Analysis Report with Action Buttons

When the user clicks **Analyze Annotations**, the page enters the analyzing state. A spinner and status text appear while the frontend parses every JSON file, extracts object IDs from each combination (bag, threats, non-threats), and compares them against the library item cache. The library IDs are pre-fetched via `/api/items/ids` (with a paginated fallback to `/api/items/categorized`) when the page loads.

The analysis report displays four summary tiles: **Total Combinations**, **Total Objects Used**, **Unique Objects**, and **Missing from System**. Below that, the **Object Availability** section breaks down objects into **Present in Library** and **Not in Library**.

Two action buttons are available at the bottom of the report:

- **Upload Missing Files** — Navigates to the Upload Page to add missing objects (shown only when invalid combinations exist)
- **Continue with Available Files** — Proceeds to rendering using only valid combinations (shown only when at least one valid combination exists)

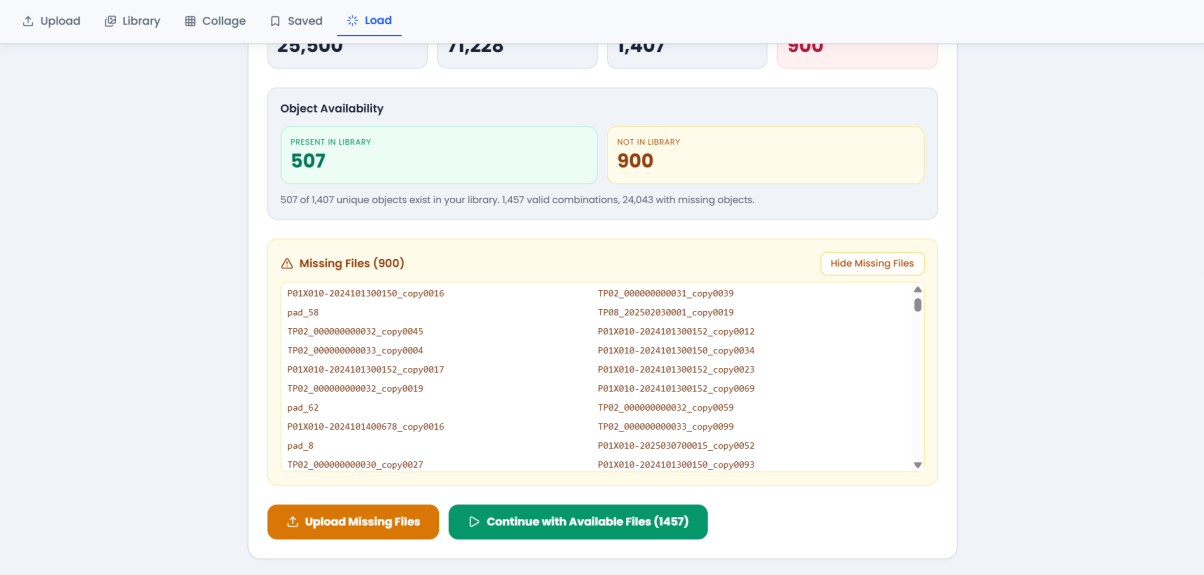


Figure 49: Missing Files List

If any objects are missing from the library, the report shows a **Missing Files** section. A toggle button lets the user **Show** or **Hide** the list of missing object IDs, making it easy to identify exactly which items need to be added.

6.2 Generating and Results

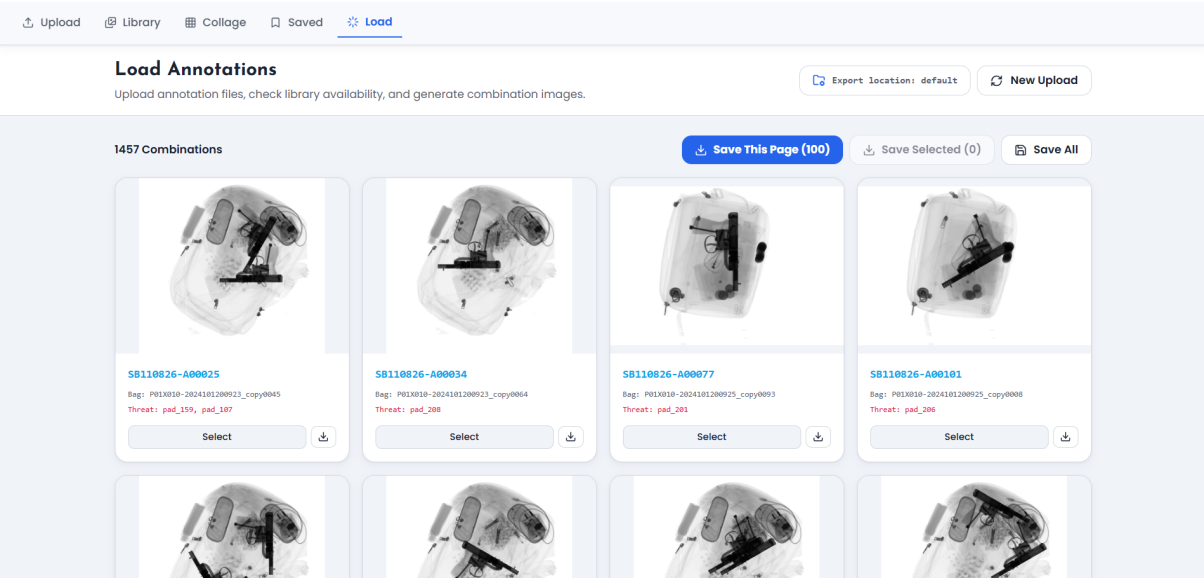


Figure 50: Results Grid with Combination Cards

When the user continues with valid combinations, the page enters the generating state. Previews for each combination are rendered via `POST /api/render-preview` with concurrency limited to

3 simultaneous renders. The page loads **100 combinations at a time** and maintains a buffer for the next page. Previews for the next page are pre-rendered once the current page finishes, so navigation is smooth.

The results view shows a responsive grid of combination cards. Each card displays the rendered preview, layout ID, bag ID, threat IDs, and other IDs. The header provides three save options:

- **Save This Page** — Exports only the 100 combinations currently visible, then advances to the next page
- **Save Selected** — Exports only the combinations the user has selected
- **Save All** — Exports every valid combination

All exports save **high- and low-resolution images** to the configured exports folder. The export location is managed via the **Export Location** selector in the header; if no custom path is set, the system uses the default load layouts folder.

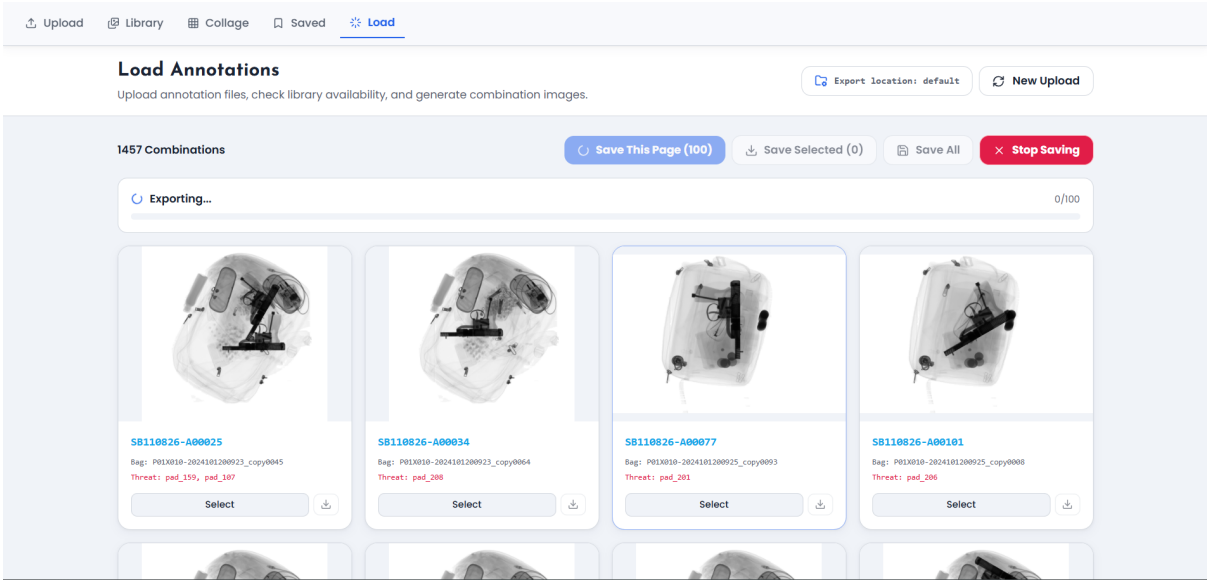


Figure 51: Export Progress Banner

During export, a progress banner appears below the toolbar. It shows an animated progress bar with a `done/total` count. The user can click **Stop Saving** to cancel the export at any time. After export completes, a toast notification confirms the number of combinations exported and the destination path.

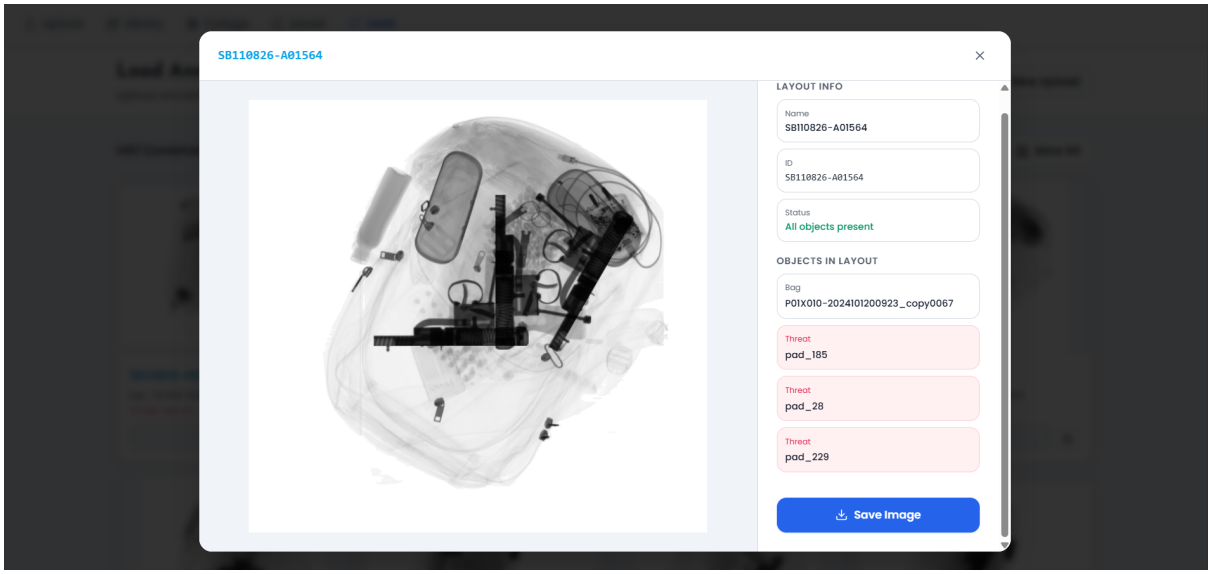


Figure 52: High-Resolution Preview Modal

Clicking any combination card opens a high-resolution preview modal. The modal is divided into two panels:

- **Left panel** — Full-size rendered image of the combination
- **Right panel** — Layout information: name, ID, status, list of objects grouped by type (Bag, Threat, Other), and any missing object IDs. A **Save Image** button at the bottom exports this single combination.

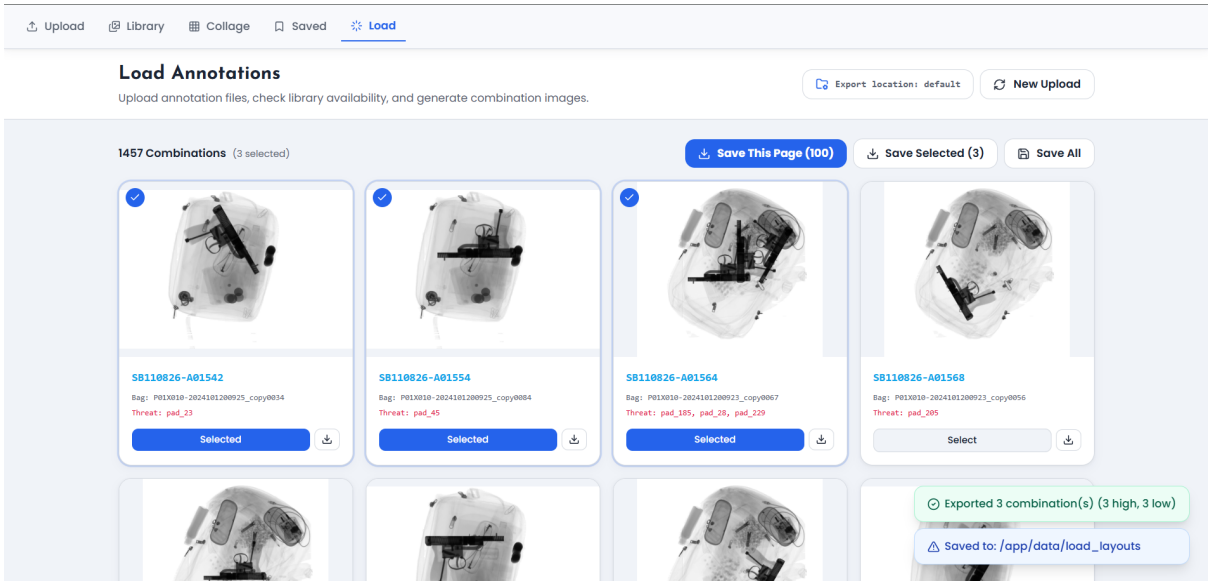


Figure 53: Selection, Save Selected, and Notification

The user can select multiple combinations by clicking the **Select** button on each card. Selected cards show an accent-colored checkmark overlay. After clicking **Save Selected** and confirming, a success toast appears showing the number of combinations exported and the folder path where the files were saved.

6.3 Key APIs

- `GET /api/items/ids` — Fetch all library item IDs for fast availability checking
- `GET /api/items/categorized` — Paginated fallback for fetching all library items
- `POST /api/render-preview` — Render a combination preview image from an objects payload
- `POST /api/load/export-combinations` — Export combinations as high + low resolution images with streaming progress
- `POST /api/load/export-combinations/cancel` — Cancel a running export job by `jobId`
- `GET /api/user-config/exports-path` — Read configured exports folder path
- `POST /api/user-config/exports-path` — Save exports folder path

6.4 Annotation Storage Format

The backend stores each saved combination as a JSON annotation file with the following structure:

Annotation file structure

```
{
  "layoutId": "SB13082619-A01001",
  "layoutName": "SB13082619-A01001",
  "createdAt": "2026-08-13T07:21:54.306Z",
  "exportedAt": "2026-08-13T07:22:12.896Z",
  "highEnergyImage": "SB_SB13082619-A01001_high.png",
  "lowEnergyImage": "SB_SB13082619-A01001_low.png",
  "objects": {
    "bag": {
      "id": "P01X010-2024101200923",
      "subcategory": "backpack",
      "filepath": "P01X010-2024101200923.png",
      "center": { "x": 352, "y": 332.5 },
      "size": { "width": 704, "height": 665 },
      "polygonOutline": [ { "x": 0.51, "y": 0.04 }, ... ],
      "validRegionPolygon": [ { "x": 713, "y": 357.9 }, ... ],
      "layer": 0,
      "tags": ["books", "board-game", "blanket", "charger"]
    },
    "threats": [
      {
        "id": "TP08_202502030001",
        "subcategory": "pistol",
        "filepath": "TP08_202502030001.png",
        "center": { "x": 379.1, "y": 228.3 },
        "size": { "width": 240, "height": 200 },
        "rotation": 0,
        "layer": 1,
        "segmentationMask": [ { "x": 0.51, "y": 0.22 }, ... ]
      }
    ],
    "non-threats": []
  }
}
```

Key points about annotation storage:

- **Layout metadata:** `layoutId`, `layoutName`, `createdAt`, and `exportedAt` identify and timestamp each record.

- **Image references:** `highEnergyImage` and `lowEnergyImage` point to the exported PNG filenames.
- **Objects:** The `objects` map contains the bag (single object), an array of `threats`, and an array of `non-threats`.
- **Polygon arrays:** `polygonOutline` and `validRegionPolygon` are arrays of normalized coordinate pairs defining the bag outline and valid placement region.
- **Segmentation masks:** Threat objects store their shape as a `segmentationMask` array of normalized coordinate pairs.
- **Layer ordering:** Each object carries a `layer` value to control rendering order.

When exported in bulk mode, multiple annotations are aggregated into partitioned files named `annotations_N.json` to avoid creating thousands of individual files.